

#1 · Wind Load Determination

What ASCE 7-16 chapters govern wind loads on buildings and other structures?

Answer

Chapters 26 through 30 provide the general requirements, exposure and topographic factors, and the analytical procedures (Directional and Envelope) used to determine wind pressures on the main wind force resisting system and components and cladding.

#2 · Wind Load Determination

Define MWFRS in wind load design.

Answer

The Main Wind Force Resisting System is the assemblage of structural elements, such as frames and shear walls, that work together to transfer wind loads from the building envelope to the foundation and resist overall lateral and overturning forces.

#3 · Wind Load Determination

How do C&C; wind pressures differ from MWFRS pressures?

Answer

Components and Cladding pressures apply to elements like windows, roof panels, and fasteners that directly receive wind and transfer it to the MWFRS. C&C; pressures are generally higher over small tributary areas due to localized peak effects.

#4 · Wind Load Determination

List the three wind Exposure Categories in ASCE 7-16.

Answer

Exposure B applies to urban and suburban areas with closely spaced obstructions. Exposure C applies to open terrain with scattered obstructions. Exposure D applies to flat unobstructed areas facing large bodies of water.

#5 · Wind Load Determination

What does the topographic factor K_{zt} account for?

Answer

K_{zt} accounts for wind speed-up effects caused by isolated hills, ridges, or escarpments that constitute abrupt changes in general topography, increasing wind speed and pressure on structures sited near the crest of such features.

#6 · Wind Load Determination

What is the purpose of the gust effect factor G ?

Answer

The gust effect factor accounts for the dynamic amplification of wind loading caused by turbulence interacting with the structure. Rigid structures use G equal to 0.85, while flexible or dynamically sensitive structures require a calculated G_f .

#7 · Wind Load Determination

Define an enclosed building per ASCE 7-16 enclosure classification.

Answer

An enclosed building has no unprotected openings large enough to be classified as open or partially enclosed. Enclosure classification affects the internal pressure coefficient GC_{pi} used when combining internal and external wind pressures.

#8 · Wind Load Determination

What defines a partially enclosed building?

Answer

A building is partially enclosed when the total area of openings in a wall receiving positive external pressure exceeds the sum of openings in the balance of the envelope by a specified ratio, producing a higher internal pressure coefficient.

#9 · Wind Load Determination

What is the velocity pressure exposure coefficient K_z used for?

Answer

K_z adjusts the basic wind speed for height above ground and exposure category, reflecting that wind velocity increases with height due to reduced surface friction, and is a direct multiplier in the velocity pressure equation q_z .

#10 · Wind Load Determination

State the basic velocity pressure equation used in ASCE 7-16.

Answer

q_z equals 0.00256 times K_z times K_{zt} times K_d times K_e times V squared, where V is the basic wind speed in mph, giving velocity pressure in psf at height z above ground for the given site conditions.

#11 · Wind Load Determination

What does the wind directionality factor K_d represent?

Answer

K_d accounts for the reduced probability of maximum winds occurring from any given direction and the reduced probability of the maximum pressure coefficient occurring simultaneously, and its value depends on the type of structure being analyzed.

#12 · Wind Load Determination

What is the ground elevation factor K_e used for in ASCE 7-16?

Answer

K_e adjusts velocity pressure for air density variation with site elevation above sea level, since denser air at lower elevations produces higher wind pressure for the same wind speed than less dense air at high elevation sites.

#13 · Wind Load Determination

Which ASCE 7-16 procedure applies to low-rise buildings meeting simplified diaphragm conditions?

Answer

The Envelope Procedure (Chapter 28) applies to enclosed and partially enclosed low-rise buildings meeting its applicability limits, using pre-tabulated external pressure coefficients ($G C_{pf}$) that already envelope the worst case across wind directions; the internal pressure coefficient ($G C_{pi}$) must still be applied separately based on enclosure classification, exactly as with the Directional Procedure.

#14 · Wind Load Determination

When must the Directional Procedure be used instead of the Envelope Procedure?

Answer

The Directional Procedure (Chapter 27) is used for buildings of any height that do not meet the low-rise limitations of the Envelope Procedure, or when the engineer needs pressures resolved by specific wind direction and surface.

#15 · Wind Load Determination

How is a building classified as rigid versus flexible for wind design?

Answer

A building is rigid if its fundamental natural frequency is 1 Hz or greater. Buildings with a fundamental frequency less than 1 Hz are classified as flexible and require the dynamic gust effect factor G_f procedure.

#16 · Wind Load Determination

What is the internal pressure coefficient GC_{pi} used for?

Answer

GC_{pi} accounts for pressure that develops inside a building through openings in the envelope, and is combined algebraically with external pressures since internal pressure can add to or subtract from the net design pressure on a surface.

#17 · Wind Load Determination

How does Risk Category affect the basic wind speed selection?

Answer

Higher Risk Category structures, such as essential facilities in Risk Category IV, are assigned higher mapped basic wind speeds corresponding to longer mean recurrence intervals, reflecting the greater consequence of failure for these occupancies.

#18 · Wind Load Determination

What is the purpose of the external pressure coefficient C_p ?

Answer

C_p reflects how wind pressure varies across different zones of a building surface, such as windward wall, leeward wall, side walls, and roof, based on wind tunnel derived data tabulated by building geometry and surface location.

#19 · Wind Load Determination

Why do roof corners and edges typically have higher C&C; pressure coefficients?

Answer

Wind separates and forms vortices at roof edges and corners, creating localized suction pressures much greater than interior roof zones. ASCE 7-16 divides roofs into zones with progressively higher negative pressure coefficients near edges and corners.

#20 · Wind Load Determination

What is a parapet's effect on roof C&C; pressures?

Answer

A continuous parapet of sufficient height reduces the negative pressure coefficients at roof edges and corners by disrupting the vortex formation that otherwise causes high suction, allowing the designer to use reduced tabulated zone pressures.

#21 · Wind Load Determination

Define effective wind area for components and cladding design.

Answer

Effective wind area is the tributary area used to determine pressure coefficients, generally taken as the span length times an effective width, but not less than one-third the span length squared, to account for load-sharing effects.

#22 · Wind Load Determination

How does effective wind area influence the C&C; pressure coefficient magnitude?

Answer

As effective wind area increases, the design pressure coefficient generally decreases because larger areas average out localized peak gust pressures, so small elements like individual fasteners experience proportionally higher design pressures than large panels.

#23 · Wind Load Determination

What building classification triggers the use of ASCE 7-16 Chapter 27 all heights method?

Answer

Buildings of any height, including those exceeding 160 feet or not meeting the regularity and diaphragm requirements of the low-rise provisions, must use the Directional Procedure general building method covering all heights.

#24 · Wind Load Determination

What is the significance of the design wind speed map in ASCE 7-16?

Answer

ASCE 7-16 provides separate wind speed maps for Risk Categories I, II, III, and IV, each showing the 3-second gust basic wind speed at 33 feet in Exposure C, corresponding to different mean recurrence intervals.

#25 · Wind Load Determination

Why is the 3-second gust wind speed used as the basic wind speed metric?

Answer

The 3-second gust duration reflects the averaging time that produces peak pressures most relevant to typical building response, replacing older fastest-mile wind speed definitions, and is standardized across all ASCE 7-16 wind speed maps.

#26 · Wind Load Determination

How is wind load combined with other loads for strength design under ASCE 7-16?

Answer

Wind load W is combined in load combinations such as $1.2D$ plus $1.0W$ plus L plus $0.5(L_r$ or S or $R)$, using strength-level wind loads already incorporating the 0.6 factor reduction embedded in the ASCE 7-16 procedures.

#27 · Wind Load Determination

What role does building height play in determining velocity pressure at a given elevation?

Answer

Velocity pressure q_z varies with height z above ground through the exposure coefficient K_z , which increases with height because wind speed increases as the boundary layer effect of ground friction diminishes further from grade.

#28 · Wind Load Determination

What distinguishes windward wall pressure from leeward wall pressure in the Directional Procedure?

Answer

Windward wall pressure is positive and varies with height using q_z , while leeward wall pressure is a constant negative (suction) pressure calculated using q_h at mean roof height, since leeward pressure does not vary significantly with height.

#29 · Wind Load Determination

What is meant by torsional load case in wind design per ASCE 7-16?

Answer

The torsional load case applies eccentric wind pressures across the building plan to capture torsional response, requiring a portion of the full design pressure applied with an eccentricity to induce twisting in addition to translation.

#30 · Wind Load Determination

How does surface roughness category affect the selection of Exposure Category?

Answer

Surface roughness is evaluated in the upwind direction for a specified distance, and the prevailing roughness in that sector, ranging from open terrain to closely spaced obstructions, determines whether Exposure B, C, or D applies to the site.

#31 · Wind Load Determination

What is the purpose of the escarpment and hill Kzt equations in ASCE 7-16?

Answer

The Kzt equations model wind speed-up as a function of hill height, horizontal distance from the crest, and building position, allowing engineers to quantify increased pressures for structures located on or near topographic features.

#32 · Wind Load Determination

Why must a rooftop-mounted air handling unit be checked separately for wind uplift?

Answer

Rooftop equipment experiences its own localized pressure coefficients per ASCE 7-16 provisions for rooftop structures and equipment, since roof-level wind flow separation creates uplift forces distinct from those on the main roof diaphragm surface below.

#33 · Wind Load Determination

What does ASCE 7-16 mean by an open building for wind classification purposes?

Answer

An open building has each wall at least 80 percent open, allowing wind to pass through with minimal obstruction, which changes the applicable internal pressure coefficient and often requires different pressure coefficients than an enclosed structure.

#34 · Wind Load Determination

Why is the along-wind response different from the across-wind response for flexible buildings?

Answer

Along-wind response results from turbulence buffeting in the direction of wind flow and is captured by the gust effect factor G_f , while across-wind (crosswind) response arises from vortex shedding, requiring separate dynamic analysis for very flexible towers.

#35 · Wind Load Determination

What information does a wind tunnel study provide that the ASCE 7-16 analytical procedure cannot?

Answer

A wind tunnel study captures site-specific effects such as shielding from adjacent buildings, complex geometry, and channeling, providing directional pressure data more accurate than tabulated analytical coefficients for unusual or tall building shapes.

#36 · Wind Load Determination

How does the minimum design wind load provision in ASCE 7-16 function?

Answer

ASCE 7-16 specifies a minimum design wind pressure on the MWFRS, typically 16 psf on wall projections and 8 psf on roof projections, ensuring a floor value even when calculated pressures from small tributary areas would be lower.

#37 · Wind Load Determination

What is the significance of mean roof height h in wind pressure calculations?

Answer

Mean roof height, the average of the eave and ridge height for pitched roofs, establishes the reference elevation for velocity pressure q_h used in leeward wall, roof, and many C&C; pressure calculations across the building envelope.

#38 · Wind Load Determination

Why does a hip roof generally experience lower wind uplift than a gable roof of similar slope?

Answer

The sloped end surfaces of a hip roof reduce flow separation and vortex intensity compared to the vertical gable end, which creates a more abrupt discontinuity in the wind stream and higher localized negative pressure coefficients.

#39 · Wind Load Determination

How should an engineer treat a canopy or overhang when computing roof wind pressures?

Answer

Overhangs experience both the roof pressure acting on the top surface and an additional pressure acting on the underside from the windward wall pressure, so both surfaces must be combined to determine the net design pressure.

#40 · Wind Load Determination

What structural consequence results from underestimating the internal pressure coefficient GCpi on an essential facility?

Answer

Underestimating GCpi understates the net pressure differential across the envelope, which can lead to undersized cladding fasteners, uplift straps, or roof diaphragm connections, risking envelope failure and progressive damage during a design-level wind event.

#41 · Seismic Load Determination

What do the mapped acceleration parameters S_s and S_1 represent?

Answer

S_s is the mapped maximum considered earthquake spectral response acceleration at short periods (0.2 second), and S_1 is the corresponding value at a 1-second period, both taken from USGS seismic hazard maps referenced in ASCE 7-16.

#42 · Seismic Load Determination

Why are S_s and S_1 modified by site coefficients F_a and F_v ?

Answer

F_a and F_v adjust the mapped bedrock spectral values for local soil conditions because softer soils amplify ground motion at certain periods, so the site-modified parameters S_{MS} and S_{M1} better represent actual expected shaking at the ground surface.

#43 · Seismic Load Determination

How are SDS and SD1 derived from SMS and SM1?

Answer

SDS equals two-thirds of SMS, and SD1 equals two-thirds of SM1. This two-thirds factor converts the maximum considered earthquake spectral accelerations to design-level spectral accelerations used for determining the Seismic Design Category and design forces.

#44 · Seismic Load Determination

What defines Site Class D under ASCE 7-16 Chapter 20?

Answer

Site Class D represents stiff soil, generally characterized by an average shear wave velocity between 600 and 1200 feet per second in the upper 100 feet of the profile, and is the default site class when insufficient data exists.

#45 · Seismic Load Determination

How is Seismic Design Category determined for a structure?

Answer

SDC is determined by taking the more severe of two classifications, one based on SDS and Risk Category and one based on SD1 and Risk Category, from tables in ASCE 7-16 Chapter 11, then selecting the higher category.

#46 · Seismic Load Determination

What is the practical significance of Seismic Design Category D or higher?

Answer

SDC D, E, or F triggers stricter detailing requirements, mandatory special seismic systems in many cases, plan and vertical irregularity limitations, and often prohibits the use of ordinary detailing categories that are otherwise permitted in lower SDCs.

#47 · Seismic Load Determination

Define the Importance Factor I_e used in seismic design.

Answer

I_e amplifies seismic design forces for structures assigned to higher Risk Categories, reflecting the greater consequence of failure for essential or high-occupancy facilities, with values of 1.0, 1.25, or 1.5 depending on Risk Category II through IV.

#48 · Seismic Load Determination

What does the response modification coefficient R represent?

Answer

R accounts for the inherent ductility and overstrength of a seismic force-resisting system, allowing design forces to be reduced from the elastic level because the system is expected to yield and dissipate energy during a design earthquake.

#49 · Seismic Load Determination

What is the deflection amplification factor C_d used for?

Answer

C_d amplifies the elastic displacements computed under reduced design seismic forces to estimate actual inelastic displacements expected during a design earthquake, which are then compared against the code-specified story drift limits.

#50 · Seismic Load Determination

Define the overstrength factor Ω_0 and its typical application.

Answer

Ω_0 represents the reserve strength beyond first yield in a seismic system, and is used to amplify forces in elements requiring overstrength design, such as collectors and connections to brittle elements, to ensure they remain elastic.

#51 · Seismic Load Determination

Summarize the general procedure of the Equivalent Lateral Force method.

Answer

The ELF procedure calculates a total seismic base shear from the design response spectrum and building weight, then distributes that base shear vertically over the building height in proportion to mass and height, approximating dynamic response with static forces.

#52 · Seismic Load Determination

State the seismic base shear equation used in the ELF procedure.

Answer

Base shear V equals C_s times W , where C_s is the seismic response coefficient derived from SDS , $SD1$, R , and I_e , and W is the effective seismic weight of the structure including applicable portions of dead and other loads.

#53 · Seismic Load Determination

What load components are typically included in effective seismic weight W ?

Answer

Effective seismic weight includes the total dead load, 25 percent of floor live load in storage occupancies, snow load exceeding 30 psf partially included, and the weight of permanent partitions and significant fixed equipment.

#54 · Seismic Load Determination

How does the vertical distribution exponent k affect seismic force allocation in the ELF method?

Answer

The exponent k , ranging from 1.0 for periods of 0.5 second or less to 2.0 for periods of 2.5 seconds or more, increases the proportion of lateral force assigned to upper stories as building period and flexibility increase.

#55 · Seismic Load Determination

What is the approximate fundamental period T_a used for in seismic design?

Answer

T_a is an empirical estimate of building period based on height and structural system type, used to determine the seismic response coefficient C_s when a more rigorous modal analysis has not been performed, bounding the calculated period.

#56 · Seismic Load Determination

Why is there an upper limit C_u on the calculated fundamental period T ?

Answer

The upper limit coefficient C_u prevents engineers from using an unrealistically long calculated period, which would understate the seismic base shear, since actual building stiffness rarely justifies periods far beyond the approximate empirical estimate T_a .

#57 · Seismic Load Determination

Define the redundancy factor ρ in ASCE 7-16 seismic design.

Answer

ρ is a penalty factor, either 1.0 or 1.3, applied to seismic force effects for structures assigned to SDC D, E, or F, reflecting whether the seismic force-resisting system has sufficient redundancy against single-element failure.

#58 · Seismic Load Determination

Under what condition can the redundancy factor ρ be taken as 1.0?

Answer

ρ equals 1.0 when the structure satisfies specific configuration requirements demonstrating adequate redundancy, such as removal of an individual element not causing more than a 33 percent reduction in story strength or creating an extreme torsional irregularity.

#59 · Seismic Load Determination

What is the seismic load effect E used in strength design load combinations?

Answer

E equals rho times QE plus or minus 0.2 times SDS times D, combining the horizontal seismic force effect QE amplified by redundancy with a vertical seismic effect that accounts for vertical ground motion acting on the dead load.

#60 · Seismic Load Determination

When is the overstrength seismic load combination Emh required instead of the standard E?

Answer

Emh, using Omega-zero times QE, is required for elements such as collectors, transfer diaphragms, and connections to discontinuous systems where brittle failure must be avoided by ensuring the element remains elastic while the ductile system yields.

#61 · Seismic Load Determination

What is meant by a vertical irregularity in seismic design?

Answer

A vertical irregularity is an abrupt discontinuity in stiffness, strength, mass, or geometry over the building height, such as a soft story or weak story, which can concentrate seismic damage and trigger additional design and analysis requirements.

#62 · Seismic Load Determination

Define a soft story irregularity under ASCE 7-16 Table 12.3-2.

Answer

A soft story exists when a story's lateral stiffness is less than 70 percent of the story above or less than 80 percent of the average stiffness of the three stories above, indicating a disproportionate concentration of drift demand.

#63 · Seismic Load Determination

What is a torsional irregularity and when is it identified?

Answer

A torsional irregularity exists when the maximum story drift at one end of a structure exceeds 1.2 times the average story drift at both ends, indicating the diaphragm and vertical elements are not symmetrically resisting lateral torsion.

#64 · Seismic Load Determination

What triggers a mandatory modal response spectrum analysis instead of the ELF procedure?

Answer

Structures with certain plan or vertical irregularities in higher SDCs, or those exceeding height and period limitations for the ELF method, must use modal response spectrum analysis to more accurately capture higher mode dynamic response effects.

#65 · Seismic Load Determination

What is the purpose of scaling modal response spectrum base shear to the ELF base shear?

Answer

ASCE 7-16 requires scaling the modal analysis base shear up to a minimum percentage of the ELF base shear (typically 100 percent for irregular structures) to prevent underestimating design forces due to modeling assumptions or period calculation errors.

#66 · Seismic Load Determination

How does Risk Category influence the level of seismic detailing required, apart from the Importance Factor?

Answer

Higher Risk Category structures often face stricter Seismic Design Category assignment thresholds since the SDC tables use lower SDS and SD1 breakpoints for Risk Category III and IV, pushing essential facilities into more demanding categories sooner.

#67 · Seismic Load Determination

What does the term seismic force-resisting system (SFRS) mean?

Answer

The SFRS is the specific structural system, such as a special moment frame or special shear wall, assigned particular R, Cd, and Omega-zero values in ASCE 7-16 Table 12.2-1, engineered to resist lateral seismic forces through prescribed ductile behavior.

#68 · Seismic Load Determination

Why do special seismic systems generally have higher R values than ordinary systems?

Answer

Special systems incorporate stricter ductile detailing requirements that allow greater inelastic deformation capacity before failure, so codes assign them higher R values, permitting a larger reduction in elastic design force compared to ordinary systems.

#69 · Seismic Load Determination

What is the height limitation purpose for seismic systems in ASCE 7-16 Table 12.2-1?

Answer

Height limits restrict the use of less ductile systems, such as ordinary moment frames, in higher Seismic Design Categories and taller buildings, since higher structures generate greater seismic demand that less ductile systems cannot reliably accommodate.

#70 · Seismic Load Determination

What is the purpose of the site-specific ground motion procedure in ASCE 7-16 Chapter 21?

Answer

Site-specific procedures develop a design response spectrum tailored to local geology, fault proximity, and soil profile using probabilistic and deterministic seismic hazard analysis, required for very soft Site Class F sites or by owner request for critical facilities.

#71 · Seismic Load Determination

How does a taller, more flexible building's fundamental period affect its seismic response coefficient C_s ?

Answer

As fundamental period increases beyond the transition period, C_s generally decreases because longer-period structures experience lower spectral acceleration demand, though a minimum C_s floor value always applies to prevent unrealistically low design forces.

#72 · Seismic Load Determination

What is the minimum value floor placed on the seismic response coefficient C_s ?

Answer

ASCE 7-16 specifies a minimum C_s of 0.044 times SDS times I_e , not less than 0.01, and for structures with S_1 of 0.6g or greater, an additional higher minimum applies to account for near-source large-magnitude effects.

#73 · Seismic Load Determination

Why does Site Class matter even for structures on identical mapped S_s and S_1 values?

Answer

Softer site classes amplify short- or long-period ground shaking differently through F_a and F_v , so two buildings on the same mapped bedrock hazard can have significantly different design spectral accelerations S_{DS} and S_{D1} depending on local soil stiffness.

#74 · Seismic Load Determination

What is meant by near-fault ground motion adjustment in seismic design?

Answer

ASCE 7-16 Chapter 21 site-specific ground motion procedures cap the deterministic MCER response spectrum near active faults; this deterministic limit, not a separate general near-fault factor applied to S_1 , is how ASCE 7-16 addresses proximity to active faults in its Chapter 11/21 procedures.

#75 · Seismic Load Determination

Define the seismic design category assignment shortcut available for structures with low SDS and SD1.

Answer

Structures assigned to SDC A, having very low seismic hazard parameters, are only required to provide a minimum lateral force equal to 1 percent of the total dead load applied at each level, without full ELF analysis.

#76 · Seismic Load Determination

What is the transition period T_L used for in ASCE 7-16 response spectra?

Answer

T_L , the long-period transition period mapped by location, marks where the design response spectrum shifts from a $1/T$ relationship to a $1/T^2$ relationship, affecting design forces for very long-period flexible structures like tall towers.

#77 · Seismic Load Determination

What does the term nonlinear response history analysis provide beyond the ELF or modal methods?

Answer

Nonlinear response history analysis directly models inelastic material and element behavior under actual or spectrum-matched ground motion time histories, capturing realistic yielding, degradation, and dynamic amplification effects that linear methods approximate through R and Cd factors.

#78 · Seismic Load Determination

How is the seismic weight for a rooftop mechanical unit typically treated in base shear calculation?

Answer

Rooftop mechanical unit weight is included in the effective seismic weight of the level it sits on for global base shear, but is also separately checked against Chapter 13 nonstructural component seismic force requirements for anchorage design.

#79 · Seismic Load Determination

What is an extreme torsional irregularity and how does it differ from a standard torsional irregularity?

Answer

An extreme torsional irregularity occurs when the maximum story drift at one end exceeds 1.4 times the average drift, a more severe threshold than the 1.2 ratio for a standard torsional irregularity, triggering additional restrictions in higher SDCs.

#80 · Seismic Load Determination

Why does ASCE 7-16 require accidental torsion to be applied even to symmetric buildings?

Answer

Accidental torsion, applied as a 5 percent offset of the center of mass from its calculated location, accounts for uncertainties in mass and stiffness distribution and construction tolerances that could induce unintended torsional response even in nominally symmetric structures.

#81 · Seismic Load Determination

What is the amplification factor A_x used for in torsional irregularity conditions?

Answer

A_x amplifies the accidental torsional moment at levels where a torsional irregularity exists, increasing the design torsion above the standard 5 percent eccentricity to better capture the disproportionate torsional response those irregular structures exhibit.

#82 · Seismic Load Determination

Define a diaphragm discontinuity irregularity in ASCE 7-16 Table 12.3-1.

Answer

A diaphragm discontinuity irregularity exists when a diaphragm has an abrupt change in stiffness, such as a large cutout or opening exceeding 50 percent of the gross enclosed diaphragm area, disrupting uniform in-plane load transfer.

#83 · Seismic Load Determination

What is an out-of-plane offset irregularity?

Answer

An out-of-plane offset irregularity occurs when a vertical lateral force-resisting element is offset horizontally from the element below it, creating a discontinuous load path that requires special design attention at the transfer level.

#84 · Seismic Load Determination

What does a nonparallel system irregularity describe?

Answer

A nonparallel system irregularity exists when vertical lateral force-resisting elements are not parallel to or symmetric about the major orthogonal axes of the seismic force-resisting system, complicating orthogonal load combination requirements for design.

#85 · Seismic Load Determination

Why are structures with a nonparallel system irregularity subject to orthogonal combination requirements?

Answer

Since the lateral system elements are not aligned with the primary axes, seismic input from one direction can generate demand in multiple element orientations simultaneously, requiring the 100 percent plus 30 percent orthogonal combination rule for adequate design.

#86 · Seismic Load Determination

What is the purpose of the weak story irregularity check under Table 12.3-2?

Answer

A weak story irregularity exists when a story's lateral strength is less than 80 percent of the story above, identifying stories where an inelastic strength deficiency could concentrate damage and cause disproportionate collapse risk during a design earthquake.

#87 · Seismic Load Determination

How does an extreme weak story differ from a weak story, and what does it prohibit?

Answer

An extreme weak story has lateral strength less than 65 percent of the story above, and per Section 12.3.3.2 structures assigned to SDC D through F with this condition are not flatly prohibited, but are limited to no more than two stories or 30 feet of structural height at that story, since the strength deficiency is considered too severe to permit taller construction.

#88 · Seismic Load Determination

What role does F_a play differently than F_v across the design spectrum?

Answer

F_a modifies the short-period mapped acceleration S_s , primarily affecting stiffer, shorter-period structures, while F_v modifies the 1-second mapped acceleration S_1 , primarily affecting more flexible, longer-period structures, since soil amplification varies by period range.

#89 · Seismic Load Determination

Why is Site Class F treated as a special case requiring site-specific analysis?

Answer

Site Class F includes soils vulnerable to potential failure or collapse under seismic loading, such as liquefiable soils, highly organic clays, or very high plasticity clays, where standard F_a and F_v tables do not reliably capture response behavior.

#90 · Seismic Load Determination

What is the general concept behind capacity design in seismic detailing?

Answer

Capacity design ensures that ductile yielding elements, like beams in a moment frame, are the intended weak link, while brittle elements such as columns and connections are designed with sufficient strength margin to remain elastic during yielding.

#91 · Seismic Load Determination

How does the strong column weak beam concept relate to capacity design in moment frames?

Answer

Strong column weak beam design ensures beam yielding occurs before column yielding at a joint, preventing a story mechanism from forming in the columns, which would be far more susceptible to collapse than distributed beam-hinge yielding.

#92 · Seismic Load Determination

What is the significance of the design spectral response acceleration parameter at the structure's fundamental period?

Answer

The spectral acceleration at the structure's own fundamental period, read from the design response spectrum, directly governs the seismic response coefficient C_s used in base shear, meaning period estimation accuracy directly affects computed design force magnitude.

#93 · Seismic Load Determination

Why might a structural engineer choose to intentionally lengthen a building's period through design?

Answer

Lengthening the fundamental period, such as through a more flexible frame system, can reduce spectral acceleration demand on the descending branch of the response spectrum, though it also increases displacement demand, requiring a balanced tradeoff evaluation.

#94 · Seismic Load Determination

What does the term seismic response coefficient C_s physically represent?

Answer

C_s represents the fraction of the building's effective seismic weight applied as the total design base shear, essentially converting the spectral acceleration demand, ductility reduction, and importance amplification into a single design coefficient for practical use.

#95 · Seismic Load Determination

How does ASCE 7-16 treat the seismic weight contribution of a permanently installed heavy equipment platform?

Answer

The weight of permanently attached equipment exceeding a minimal threshold must be included in the effective seismic weight W , since this mass participates in the building's dynamic response and directly influences the base shear magnitude.

#96 · Seismic Load Determination

What is the practical difference between Risk Category II and Risk Category III structures for seismic design?

Answer

Risk Category III structures, representing substantial public hazard occupancies, receive an Importance Factor of 1.25 versus 1.0 for Risk Category II, increasing design seismic forces by 25 percent to reduce damage likelihood for higher-consequence occupancies.

#97 · Seismic Load Determination

What triggers the requirement to design collector elements for overstrength seismic forces?

Answer

Collectors gathering diaphragm shear into discrete vertical lateral force-resisting elements must generally be designed for the overstrength seismic load combination using Omega-zero, since a brittle collector failure could prevent the ductile system from developing its full capacity.

#98 · Seismic Load Determination

How is the seismic base shear formula affected when SD1 governs rather than SDS?

Answer

When the calculated period exceeds the short-period plateau, C_s is instead computed from SD1 divided by T times R over I_e , meaning longer-period structures have their base shear governed by the long-period spectral parameter rather than the short-period plateau.

#99 · Seismic Load Determination

What is meant by the term seismic hazard exposure when comparing two sites with identical Site Class but different locations?

Answer

Even with the same Site Class, two locations can have very different mapped S_s and S_1 values reflecting proximity to active faults and regional seismicity, meaning geographic location remains the primary driver of underlying seismic hazard exposure.

#100 · Seismic Load Determination

Why does ASCE 7-16 require a minimum base shear even for very long-period, highly flexible structures?

Answer

The minimum base shear floor prevents extremely flexible structures from being designed for unrealistically small forces due to the inverse relationship between period and spectral acceleration, ensuring baseline strength and stability against seismic and other lateral demands.

#101 · Seismic Load Determination

What is the purpose of orthogonal load combination requirements in higher Seismic Design Categories?

Answer

Orthogonal combination requires combining 100 percent of seismic force in one direction with 30 percent in the perpendicular direction, since earthquake ground motion is multidirectional and elements at building corners can experience simultaneous demand from both principal directions.

#102 · Seismic Load Determination

How does the number of stories interact with SDC to determine allowable seismic force-resisting systems?

Answer

ASCE 7-16 Table 12.2-1 imposes structural height limits that vary by system type and SDC, so a taller building in SDC D may be restricted to only the most ductile special systems while shorter buildings retain more system options.

#103 · Seismic Load Determination

What is the seismic design implication of assigning a structure to Risk Category IV?

Answer

Risk Category IV structures, considered essential facilities, receive the highest Importance Factor of 1.5, face the most stringent SDC assignment thresholds, and are subject to enhanced nonstructural component anchorage and continued-operation performance expectations.

#104 · Seismic Load Determination

Why is the vertical seismic load effect 0.2 times SDS times D included in the load combination?

Answer

This term approximates the effect of vertical ground acceleration acting on the structure's dead load, which can reduce effective gravity resistance in uplift-sensitive members or increase compression demand, and is added or subtracted depending on the governing case.

#105 · Seismic Load Determination

What analytical tool captures the contribution of multiple vibration modes to total seismic response?

Answer

Modal response spectrum analysis combines the response of several vibration modes, typically using square root of sum of squares or complete quadratic combination methods, to capture higher mode effects that a single-mode ELF approximation cannot represent.

#106 · Seismic Load Determination

How does soil-structure interaction potentially reduce design seismic demand?

Answer

Soil-structure interaction procedures in ASCE 7-16 Chapter 19 can reduce the effective base shear and modify the effective period by accounting for foundation flexibility and radiation damping, though application requires meeting specific eligibility criteria.

#107 · Seismic Load Determination

What is the significance of designing a diaphragm as part of the seismic force-resisting system rather than only gravity system?

Answer

Diaphragms transfer lateral seismic inertial forces from each floor level to the vertical elements, so diaphragm design must address in-plane shear, chord forces, and collector forces in addition to any gravity load-carrying function of the floor system.

#108 · Seismic Load Determination

Why does redundancy factor rho apply only to horizontal seismic force effects and not vertical seismic effects?

Answer

Rho specifically penalizes designs lacking redundant load paths for the horizontal seismic force component Q_E , since redundancy concerns relate to lateral system configuration, whereas the vertical seismic term $0.2SDS D$ is a separate gravity-related adjustment unaffected by system redundancy.

#109 · Seismic Load Determination

What does the term design earthquake ground motion mean relative to maximum considered earthquake ground motion?

Answer

Design earthquake ground motion is two-thirds of the maximum considered earthquake ground motion, reflecting a target performance level of collapse prevention with margin against the rarer, more intense maximum considered event used for hazard mapping.

#110 · Seismic Load Determination

How does ASCE 7-16 handle seismic design for structures supported on a mat foundation over soft soil?

Answer

Soft soil conditions typically raise F_a and F_v site coefficients, increasing SDS and $SD1$, while also potentially requiring liquefaction evaluation and soil-structure interaction consideration, both of which can significantly influence the mat foundation's lateral and bearing design.

#111 · Lateral Force Distribution and Analysis Methods

Define a rigid diaphragm and its effect on lateral force distribution.

Answer

A rigid diaphragm distributes lateral force to vertical elements in proportion to their relative stiffness, and can also transfer torsional forces, since the diaphragm itself does not deform significantly compared to the vertical resisting elements.

#112 · Lateral Force Distribution and Analysis Methods

Define a flexible diaphragm and its effect on lateral force distribution.

Answer

A flexible diaphragm distributes lateral force to vertical elements based on tributary area rather than relative stiffness, behaving like a simple beam spanning between supports, and generally cannot reliably transfer torsional force between elements.

#113 · Lateral Force Distribution and Analysis Methods

What is a semi-rigid diaphragm and when must it be modeled explicitly?

Answer

A semi-rigid diaphragm has stiffness between rigid and flexible extremes, and ASCE 7-16 requires explicit modeling of diaphragm flexibility when the diaphragm cannot reasonably be idealized as either fully rigid or fully flexible for accurate load distribution.

#114 · Lateral Force Distribution and Analysis Methods

What is inherent torsion in lateral force distribution?

Answer

Inherent torsion results from the actual eccentricity between the center of mass and center of rigidity of a story, causing a rigid diaphragm to rotate as it translates under lateral load, distributing additional force to elements farther from the center of rigidity.

#115 • Lateral Force Distribution and Analysis Methods

What is accidental torsion and why is it required in addition to inherent torsion?

Answer

Accidental torsion is an artificial eccentricity, typically 5 percent of the building dimension, added to the calculated center of mass to account for uncertainties in mass and stiffness distribution not captured by the nominal inherent torsion calculation.

#116 • Lateral Force Distribution and Analysis Methods

How is relative stiffness of vertical lateral elements typically determined for rigid diaphragm distribution?

Answer

Relative stiffness is determined by calculating each element's lateral stiffness, often as the inverse of its deflection under a unit load considering shear and flexural components, then apportioning story shear to each element proportional to its stiffness fraction.

#117 · Lateral Force Distribution and Analysis Methods

What is the center of rigidity in a story with multiple lateral force-resisting elements?

Answer

The center of rigidity is the point through which a lateral force can be applied without causing the rigid diaphragm to twist, calculated as the stiffness-weighted centroid of all vertical lateral force-resisting elements at that story.

#118 · Lateral Force Distribution and Analysis Methods

Why does story drift need to be checked against code limits in seismic design?

Answer

Excessive story drift can cause damage to nonstructural components, create instability from P-delta effects, and compromise structural integrity, so ASCE 7-16 Table 12.12-1 sets allowable drift limits based on Risk Category and structural system type.

#119 · Lateral Force Distribution and Analysis Methods

What is the typical allowable story drift limit for a Risk Category II building without special masonry?

Answer

For most Risk Category II structures other than masonry shear wall buildings, the allowable story drift is typically 0.020 times the story height, though masonry cantilever shear wall structures are limited to a smaller 0.010 times story height.

#120 · Lateral Force Distribution and Analysis Methods

How is design story drift calculated from the elastic analysis displacement?

Answer

Design story drift is computed by multiplying the elastic displacement from the reduced design seismic force by the deflection amplification factor C_d , then dividing by the Importance Factor I_e , to estimate actual expected inelastic drift.

#121 · Lateral Force Distribution and Analysis Methods

Define the P-delta effect in lateral force analysis.

Answer

The P-delta effect is the secondary increase in overturning moment and story shear caused by gravity loads acting through the lateral displacement of the structure, which can significantly amplify demand in flexible or heavily loaded structures.

#122 · Lateral Force Distribution and Analysis Methods

What is the stability coefficient theta used for in P-delta evaluation?

Answer

Theta quantifies the ratio of secondary moment to primary moment at a story, and ASCE 7-16 requires P-delta effects to be considered in analysis whenever theta exceeds 0.10, with an upper limit theta-max beyond which the structure is potentially unstable.

#123 · Lateral Force Distribution and Analysis Methods

What defines the load path concept in lateral force-resisting design?

Answer

Load path refers to the continuous, uninterrupted route by which lateral forces travel from their point of application at the building envelope or floor mass, through diaphragms and collectors, into vertical elements, and ultimately down to the foundation.

#124 · Lateral Force Distribution and Analysis Methods

Why is a complete and continuous load path considered the most fundamental requirement in lateral design?

Answer

Without a continuous load path, forces cannot reliably transfer between structural elements regardless of individual element strength, so ASCE 7-16 explicitly requires a complete load path capable of transmitting all applicable loads from origin to final point of resistance.

#125 · Lateral Force Distribution and Analysis Methods

What is the general distinction between the Equivalent Lateral Force method and modal response spectrum analysis?

Answer

The ELF method uses a simplified static approximation assuming a single dominant vibration mode, while modal response spectrum analysis considers the contribution of multiple vibration modes, providing more accurate results for irregular or dynamically complex structures.

#126 · Lateral Force Distribution and Analysis Methods

When is linear response history analysis preferred over modal response spectrum analysis?

Answer

Linear response history analysis directly integrates the structure's response to specific ground motion time histories rather than a smoothed design spectrum, offering more detailed time-varying response information, often used for structures with unusual dynamic characteristics or base isolation.

#127 · Lateral Force Distribution and Analysis Methods

How does building torsional stiffness affect distribution of lateral force among resisting elements?

Answer

Greater torsional stiffness, achieved by placing lateral elements farther from the center of rigidity, reduces rotational displacement demand per unit torque, more evenly distributing lateral force and reducing the additional shear imposed on elements far from the center of rigidity.

#128 · Lateral Force Distribution and Analysis Methods

What analysis consideration is needed when a diaphragm has significantly different stiffness zones, such as a large reentrant corner?

Answer

A diaphragm with abrupt stiffness changes or large reentrant corners may require finite element or semi-rigid modeling to accurately capture stress concentrations and force redistribution, since simplified rigid or flexible assumptions could misrepresent actual behavior.

#129 · Lateral Force Distribution and Analysis Methods

What is meant by a collector or drag strut element in the load path?

Answer

A collector, also called a drag strut, gathers diaphragm shear along a line and delivers it to a discrete vertical lateral force-resisting element, effectively dragging distributed diaphragm force into a concentrated connection point.

#130 · Lateral Force Distribution and Analysis Methods

How do relative stiffness calculations differ between a flexible diaphragm building and a rigid diaphragm building with the same wall layout?

Answer

In a flexible diaphragm building, wall forces depend only on tributary area regardless of wall stiffness, while in a rigid diaphragm building, a stiffer wall attracts proportionally more force even with identical tributary area, changing the design demand on each wall.

#131 • Lateral Force Distribution and Analysis Methods

What is the purpose of an envelope of forces from both flexible and rigid diaphragm assumptions in a semi-rigid analysis?

Answer

When diaphragm behavior is uncertain, designers may bound the true response by enveloping results from both flexible and rigid diaphragm idealizations, ensuring elements are designed for the higher demand predicted by either simplified model.

#132 • Lateral Force Distribution and Analysis Methods

Define torsional irregularity's effect on required analysis method selection.

Answer

A significant torsional irregularity in higher Seismic Design Categories can trigger the requirement for a more rigorous analysis method, such as modal response spectrum analysis, since the ELF method's assumptions do not adequately capture concentrated torsional response.

#133 · Lateral Force Distribution and Analysis Methods

What is the significance of chord forces in diaphragm design?

Answer

Chord forces develop at diaphragm boundaries acting like the flanges of a horizontal beam, resisting the bending moment induced by lateral load, and must be designed as tension and compression elements, often using the boundary framing or dedicated reinforcement.

#134 · Lateral Force Distribution and Analysis Methods

How does an increase in a structure's fundamental period generally affect story drift?

Answer

More flexible structures with longer fundamental periods generally experience greater lateral displacement under a given seismic force, increasing story drift, which is why drift limits often govern the design of taller or more slender lateral systems.

#135 · Lateral Force Distribution and Analysis Methods

What is meant by relative rigidity method versus tributary area method for distributing lateral force?

Answer

The relative rigidity method distributes force proportional to element stiffness for rigid diaphragms, while the tributary area method distributes force based on the area each element supports, applicable to flexible diaphragms where stiffness compatibility does not govern.

#136 · Lateral Force Distribution and Analysis Methods

Why must P-delta effects be evaluated using the amplified Cd-level displacement rather than the elastic design-level displacement?

Answer

The stability coefficient θ uses actual expected inelastic drift, which is the elastic displacement amplified by C_d , since the true secondary moment amplification depends on realistic displacement magnitude, not the artificially reduced elastic design displacement.

#137 · Lateral Force Distribution and Analysis Methods

What structural configuration commonly creates a torsional irregularity even without significant mass eccentricity?

Answer

An asymmetric arrangement of lateral force-resisting elements, such as shear walls concentrated on one side of a building with an open storefront on the opposite side, creates an eccentric center of rigidity relative to center of mass, inducing torsional irregularity.

#138 · Lateral Force Distribution and Analysis Methods

What analysis method must generally be used for structures exceeding the height limits for ELF eligibility in higher Seismic Design Categories?

Answer

Structures exceeding ELF height and regularity eligibility limits in SDC D, E, or F generally require modal response spectrum analysis, which better captures the multi-mode dynamic behavior expected in taller or more irregular structures.

#139 · Lateral Force Distribution and Analysis Methods

What is a redundancy consideration relevant to distribution of story shear among vertical elements?

Answer

Redundant systems with multiple independent lateral force-resisting elements allow load redistribution if one element is damaged or overloaded, which is why the redundancy factor ρ penalizes systems relying on very few elements to resist total story shear.

#140 · Lateral Force Distribution and Analysis Methods

How does relative element stiffness change when a shear wall has a large door or window opening?

Answer

A large opening reduces a shear wall's effective stiffness by reducing available shear and flexural resisting area, which lowers the proportion of rigid diaphragm force it attracts relative to solid walls, requiring recalculation of the stiffness distribution.

#141 · Lateral Force Distribution and Analysis Methods

What does the term drift ratio mean in the context of story drift limit checks?

Answer

Drift ratio is the story drift divided by the story height, expressed as a decimal or percentage, and is directly compared against the code-prescribed allowable drift ratio, such as 0.020, to confirm the lateral system meets serviceability and stability criteria.

#142 · Lateral Force Distribution and Analysis Methods

Why is torsional response particularly critical at building corners for element design?

Answer

Corner elements experience the combined effect of translational shear plus the maximum torsional contribution since they are farthest from the center of rigidity, often governing the peak design demand compared to elements located closer to the building's rotational center.

#143 · Lateral Force Distribution and Analysis Methods

What is the general effect of increasing R on story drift results for the same building configuration?

Answer

Increasing R reduces the elastic design force and thus the elastic displacement from analysis, but since drift is amplified by C_d , which typically scales with R, actual expected inelastic drift does not decrease proportionally and must still be separately checked.

#144 · Lateral Force Distribution and Analysis Methods

What is meant by story shear in the context of lateral load analysis?

Answer

Story shear is the total lateral force that must be resisted at a given story, equal to the sum of all lateral forces applied above that level, and is distributed among the vertical lateral force-resisting elements at that story.

#145 • Lateral Force Distribution and Analysis Methods

How does overturning moment develop differently at the base compared to at an intermediate story?

Answer

Overturning moment accumulates from the story shears and their respective moment arms above each level, so the base experiences the full cumulative overturning effect of the entire building height, while intermediate stories see only the moment from levels above them.

#146 • Lateral Force Distribution and Analysis Methods

What analysis distinction exists between story drift and total building lateral displacement?

Answer

Story drift is the relative displacement between two adjacent floor levels, while total lateral displacement is the cumulative absolute displacement at a given level measured from the base, and code drift limits apply specifically to the relative story value.

#147 · Lateral Force Distribution and Analysis Methods

What is the effect of a rigid diaphragm assumption on force sharing between a stiff concrete wall and a flexible wood shear wall in the same line?

Answer

Under a rigid diaphragm assumption, the stiffer concrete wall would attract a disproportionately large share of the lateral force compared to the flexible wood wall, which can be an unrealistic and potentially unsafe assumption if diaphragm flexibility actually limits this transfer.

#148 · Lateral Force Distribution and Analysis Methods

Why does ASCE 7-16 require a minimum accidental torsion offset even when calculated inherent eccentricity is very small?

Answer

The accidental torsion requirement is independent of the calculated inherent eccentricity because it represents unmodeled uncertainty in mass and stiffness location, so it applies as an additive minimum offset regardless of how small the nominal inherent eccentricity is calculated to be.

#149 · Lateral Force Distribution and Analysis Methods

What structural response characteristic distinguishes a torsionally flexible building from a torsionally stiff building?

Answer

A torsionally flexible building has a relatively low ratio of torsional stiffness to translational stiffness, resulting in large rotational displacements relative to translational displacement, often identified through a torsional irregularity check using story drift ratios.

#150 · Lateral Force Distribution and Analysis Methods

What is the significance of comparing static ELF results against a dynamic modal response spectrum analysis for the same structure?

Answer

Comparing both methods helps verify that the simplified ELF static approximation reasonably captures the dynamic behavior, and code-required scaling ensures the more detailed modal results are not allowed to significantly understate the simpler ELF baseline demand.

#151 · Lateral Force Distribution and Analysis Methods

How does increasing a shear wall's length typically affect its relative stiffness and force attraction under rigid diaphragm distribution?

Answer

Increasing shear wall length generally increases both flexural and shear stiffness, so a longer wall attracts a greater proportion of the total story shear under rigid diaphragm distribution compared to shorter walls of similar construction in the same line.

#152 · Lateral Force Distribution and Analysis Methods

What is the practical design consequence of a discontinuous shear wall that does not extend to the foundation?

Answer

A discontinuous shear wall requires a transfer diaphragm or transfer beam to redirect the wall's force into elements below, and this transfer element is typically designed for overstrength seismic forces due to the vertical irregularity it creates.

#153 • Lateral Force Distribution and Analysis Methods

What does the equivalent lateral force procedure assume about mode shape when distributing base shear vertically?

Answer

The ELF procedure assumes an idealized mode shape approximated by the height-weighted mass distribution using the exponent k , effectively assuming the fundamental mode governs response and higher mode contributions are negligible for regular, low-to-moderate height structures.

#154 • Lateral Force Distribution and Analysis Methods

Why can a flexible diaphragm sometimes be conservatively idealized as simply supported beams between wall lines?

Answer

A flexible diaphragm's own deflection dominates over the deflection of the supporting walls, so it behaves essentially as a simply supported beam distributing tributary load to walls regardless of relative wall stiffness, simplifying the distribution calculation considerably.

#155 · Lateral Force Distribution and Analysis Methods

What is the role of a diaphragm's shear capacity check versus its chord capacity check?

Answer

The shear capacity check confirms the diaphragm sheathing and fastening can transfer in-plane shear without failure, while the chord capacity check ensures the boundary elements can resist the tension and compression forces from diaphragm bending, both being independent design requirements.

#156 · Lateral Force Distribution and Analysis Methods

How does the concept of stiffness compatibility apply when combining different lateral system types on the same line of resistance?

Answer

Combining systems of very different stiffness and ductility, such as a rigid concrete wall alongside a flexible steel frame on the same line, can cause the stiffer element to yield first and attract disproportionate demand, complicating reliable capacity design.

#157 · Lateral Force Distribution and Analysis Methods

What is the significance of the term negative torsion reduction not being permitted under ASCE 7-16?

Answer

ASCE 7-16 does not allow reducing element forces below the translational-only value even when torsion would theoretically reduce demand on one side, since actual dynamic torsional response can reverse direction, so torsion is only permitted to add to, never subtract from, translational shear.

#158 · Lateral Force Distribution and Analysis Methods

Why is a P-delta analysis particularly important for a slender high-rise building compared to a low, stiff building?

Answer

A slender high-rise building experiences larger lateral displacements and carries substantial gravity load through a tall, flexible frame, magnifying the secondary P-delta moment, while a low, stiff building has small displacements that produce negligible secondary effects.

#159 · Lateral Force Distribution and Analysis Methods

What is meant by the term dynamic amplification in the context of story shear distribution using higher-mode participation?

Answer

Dynamic amplification refers to increased force demand at certain stories, often upper levels, resulting from higher vibration mode participation not captured by the simplified single-mode ELF vertical distribution, which modal response spectrum analysis explicitly accounts for.

#160 · Lateral Force Distribution and Analysis Methods

How is the accidental torsional moment magnitude calculated at a given floor level?

Answer

The accidental torsional moment equals the story shear at that level multiplied by the accidental eccentricity, typically 5 percent of the building dimension perpendicular to the direction of applied force, applied in the direction that produces the most severe load effect.

#161 • Lateral Force Distribution and Analysis Methods

What condition would require a structure to use the equivalent lateral force method combined with an amplified accidental torsion factor A_x ?

Answer

A structure with a torsional or extreme torsional irregularity assigned to SDC C or higher must apply the amplification factor A_x to accidental torsion, even while otherwise using the standard ELF method, unless a more rigorous dynamic analysis is instead performed.

#162 • Lateral Force Distribution and Analysis Methods

What is the fundamental reason story drift limits are more restrictive for unreinforced masonry structures than for steel moment frames?

Answer

Unreinforced masonry is brittle with very limited ductility and displacement capacity before cracking or failure, so codes impose tighter drift limits to keep deformation demand well within the material's limited tolerance compared to more ductile steel systems.

#163 · Lateral Force Distribution and Analysis Methods

Why does the seismic force-resisting system's ductility level factor into acceptable story drift limits?

Answer

More ductile systems can tolerate larger inelastic deformations without losing gravity load-carrying capacity, so ASCE 7-16 permits more lenient drift limits for ductile systems while requiring stricter limits for systems with limited post-yield deformation capacity.

#164 · Lateral Force Distribution and Analysis Methods

What is the purpose of comparing story stiffness ratios in adjacent stories when checking for vertical irregularities?

Answer

Comparing adjacent story stiffness ratios identifies abrupt stiffness discontinuities, such as a soft story, that can concentrate deformation demand and inelastic damage at a single level rather than distributing ductile behavior more uniformly through the building height.

#165 · Lateral Force Distribution and Analysis Methods

How does the presence of a rigid diaphragm change the effect of an eccentric mass distribution on element design forces?

Answer

With a rigid diaphragm, an eccentric mass distribution generates inherent torsion that adds shear demand to elements on the far side from the center of rigidity, an effect that would not occur under a flexible diaphragm relying solely on tributary area.

#166 · Lateral Force Distribution and Analysis Methods

What is the significance of checking both the +5 percent and -5 percent accidental eccentricity cases in a rigid diaphragm analysis?

Answer

Checking both eccentricity directions ensures the worst-case torsional demand is captured for each individual element, since shifting the assumed mass location in one direction may increase demand on certain elements while decreasing it on others in the same story.

#167 · Lateral Force Distribution and Analysis Methods

What analytical challenge arises when a building has multiple diaphragm levels with differing rigidity classifications?

Answer

A building may have a rigid diaphragm at one level, such as a concrete podium slab, and a flexible diaphragm at another, such as a wood-framed roof, requiring separate distribution methods to be applied consistently at each respective level during analysis.

#168 · Lateral Force Distribution and Analysis Methods

Why is the load path through a diaphragm opening around a stairwell a common design concern?

Answer

A stairwell opening interrupts the diaphragm's continuous shear transfer path, requiring boundary members around the opening designed as subchords and drag struts to reroute forces around the discontinuity while maintaining overall diaphragm integrity.

#169 · Lateral Force Distribution and Analysis Methods

What is the general relationship between building torsional radius and the severity of torsional response?

Answer

A smaller torsional radius relative to the eccentricity between center of mass and center of rigidity indicates a torsionally flexible structure that will experience more severe rotational response and greater force amplification at elements farthest from the center of rigidity.

#170 · Lateral Force Distribution and Analysis Methods

How does the redundancy factor ρ interact with the analysis method selected for determining seismic force distribution?

Answer

ρ is applied as a multiplier to the horizontal seismic force effect regardless of whether the ELF or modal response spectrum method generated that force, meaning the redundancy penalty is layered onto the distributed element forces from either analysis approach.

#171 · Lateral Force Distribution and Analysis Methods

What is a common mistake when applying tributary area distribution to a flexible diaphragm with irregular wall spacing?

Answer

A common mistake is assuming uniform tributary width along the diaphragm length rather than correctly apportioning tributary area based on the actual midpoint distances between adjacent walls, which can significantly misallocate diaphragm reaction forces to each wall line.

#172 · Lateral Force Distribution and Analysis Methods

Why is verification of computer model results against hand-calculated approximate values a required engineering practice?

Answer

Hand-check verification confirms that computer model input assumptions, boundary conditions, and load application are reasonable, since a modeling error such as an incorrect diaphragm rigidity assignment can produce plausible-looking but significantly incorrect force distribution results.

#173 • Lateral Force Distribution and Analysis Methods

What does an approximate hand calculation of base shear typically compare against when verifying a computer model?

Answer

An approximate hand calculation of total base shear using the ELF formula with estimated building weight and period is typically compared against the computer model's reported base shear to confirm the model's total applied lateral force is reasonable before trusting distributed results.

#174 • Lateral Force Distribution and Analysis Methods

How does a rigid diaphragm assumption affect the design of a collector connecting a shear wall to the diaphragm?

Answer

Under a rigid diaphragm, the collector must transfer the wall's full share of story shear as determined by relative stiffness distribution, which may be significantly different from a simple tributary area estimate, directly affecting the required collector force and connection design.

#175 · Lateral Force Distribution and Analysis Methods

What is the design significance of a torsional irregularity when selecting between the ELF procedure and modal response spectrum analysis?

Answer

A significant torsional irregularity in a structure assigned to certain Seismic Design Categories can disqualify the use of the ELF procedure alone, requiring modal response spectrum analysis to more reliably capture the coupled translational-torsional dynamic response.

#176 · Lateral Force Distribution and Analysis Methods

Why can P-delta effects lead to a story mechanism if not properly accounted for in design?

Answer

Unaddressed P-delta amplification can push a story's actual demand beyond its designed capacity as displacement grows, potentially triggering a soft or weak story mechanism where deformation concentrates disproportionately at one level, risking progressive collapse.

#177 · Lateral Force Distribution and Analysis Methods

What is the relationship between diaphragm span-to-depth ratio and the appropriateness of a flexible diaphragm assumption?

Answer

A diaphragm with a large span-to-depth ratio tends to behave more flexibly, since bending deformation becomes more significant relative to shear deformation, making the flexible diaphragm idealization more representative of actual force distribution behavior for such proportions.

#178 · Lateral Force Distribution and Analysis Methods

How should an engineer address a lateral system with vertical elements of significantly different stiffness sharing a single rigid diaphragm level?

Answer

The engineer must calculate accurate relative stiffness values for each dissimilar element type, accounting for shear and flexural deformation appropriately for each material and configuration, then distribute story shear proportionally rather than assuming equal sharing.

#179 · Lateral Force Distribution and Analysis Methods

What is the purpose of reviewing story drift results separately for each principal orthogonal direction of a building?

Answer

Lateral stiffness and configuration often differ between the two principal directions, so story drift must be independently checked in each direction against the applicable allowable limit, since satisfying the limit in one direction does not guarantee compliance in the other.

#180 · Lateral Force Distribution and Analysis Methods

Why is understanding both the analysis method assumptions and the physical load path essential before finalizing a lateral force-resisting design?

Answer

An analysis method's output is only as reliable as its underlying assumptions about diaphragm behavior, torsion, and dynamic response, so an engineer must verify the physical load path matches those assumptions to avoid a design that looks correct numerically but is not actually constructible or safe.

#181 · Steel Lateral Force-Resisting Systems

What does SMF stand for in steel lateral force-resisting systems?

Answer

Special Moment Frame. SMF provides the highest ductility of the three moment frame categories per AISC 341, using beam-column connections qualified by cyclic testing to sustain large inelastic story drift angles without significant strength loss.

#182 · Steel Lateral Force-Resisting Systems

Which AISC standard governs seismic detailing requirements for steel lateral systems?

Answer

AISC 341, Seismic Provisions for Structural Steel Buildings. It specifies ductility detailing, protected zones, demand critical welds, and connection qualification requirements for SMF, IMF, OMF, SCBF, OCBF, and EBF systems.

#183 · Steel Lateral Force-Resisting Systems

Define a protected zone in AISC 341 steel moment frame detailing.

Answer

A protected zone is the region of a member expected to undergo significant inelastic strain during a design earthquake. Welded attachments, shear studs, and discontinuities are restricted there to avoid initiating fracture at the location of highest ductility demand.

#184 · Steel Lateral Force-Resisting Systems

What is the primary purpose of capacity design in steel lateral systems?

Answer

Capacity design ensures that brittle or non-ductile elements, such as columns and connections, remain elastic while a designated ductile element yields first, forcing inelastic behavior into a predictable, detailed location rather than an undetailed brittle failure.

#185 · Steel Lateral Force-Resisting Systems

What is a key structural difference between SCBF and OCBF?

Answer

Special Concentrically Braced Frames require ductile brace and gusset plate detailing to sustain large inelastic cyclic deformation, while Ordinary Concentrically Braced Frames have minimal ductility detailing and are assigned a lower response modification coefficient R .

#186 · Steel Lateral Force-Resisting Systems

What defines an Eccentrically Braced Frame (EBF)?

Answer

An EBF intentionally offsets the brace-to-beam connection from the column or from the opposing brace, creating a short beam segment called the link that yields in shear or flexure, providing ductility while braces and columns remain essentially elastic.

#187 · Steel Lateral Force-Resisting Systems

What is the function of the link in an eccentrically braced frame?

Answer

The link is the designated ductile fuse of an EBF. It is designed to yield in shear or flexure during a design earthquake, dissipating seismic energy while the braces, columns, and remaining beam segments are capacity designed to remain essentially elastic.

#188 · Steel Lateral Force-Resisting Systems

How does an Intermediate Moment Frame (IMF) compare to an SMF in ductility?

Answer

An IMF provides moderate ductility with less stringent connection qualification and detailing requirements than an SMF, resulting in a lower response modification coefficient R and a correspondingly higher design seismic force compared to an equivalent SMF.

#189 · Steel Lateral Force-Resisting Systems

What is demand critical welding in steel seismic connections?

Answer

Demand critical welds are complete joint penetration groove welds at locations of high inelastic demand, such as beam flange to column flange connections in SMF systems, requiring enhanced notch toughness filler metal and stringent quality control per AISC 341.

#190 · Steel Lateral Force-Resisting Systems

What is the purpose of a reduced beam section (RBS) connection in an SMF?

Answer

An RBS connection trims the beam flanges near the beam-column joint to force plastic hinge formation away from the column face, reducing demand on the connection weld while still allowing the beam to develop its plastic moment capacity reliably.

#191 · Steel Lateral Force-Resisting Systems

What is an OMF and where is it typically permitted?

Answer

An Ordinary Moment Frame has minimal seismic ductility detailing requirements and the lowest R value among steel moment frame categories, typically permitted only in lower Seismic Design Categories or for structures with limited height per ASCE 7-16 Table 12.2-1.

#192 · Steel Lateral Force-Resisting Systems

Why must gusset plates in SCBF systems accommodate brace buckling?

Answer

SCBF braces are expected to buckle in compression during a design earthquake. Gusset plates must be detailed with sufficient clearance, typically using a linear or elliptical clearance zone, to allow the brace to rotate and form a plastic hinge without fracturing the plate.

#193 · Steel Lateral Force-Resisting Systems

What column requirement applies at the base of SCBF braced bays?

Answer

Columns in SCBF systems must be designed using capacity design forces derived from the expected brace strength in tension and post-buckling compression, ensuring the column remains elastic and does not fail before the ductile brace fuses activate.

#194 · Steel Lateral Force-Resisting Systems

What is the strong column weak beam requirement in SMF design?

Answer

AISC 341 requires the sum of column flexural strengths at a joint to exceed the sum of beam flexural strengths by a specified ratio, ensuring plastic hinges form in beams rather than columns, which preserves the overall lateral stability of the frame.

#195 · Steel Lateral Force-Resisting Systems

What panel zone strength check is required in SMF connection design?

Answer

The column web panel zone at each beam-column joint must be checked for adequate shear strength, using doubler plates if needed, to prevent excessive panel zone yielding that could concentrate inelastic deformation away from the intended beam hinge location.

#196 · Steel Lateral Force-Resisting Systems

How does the AISC Seismic Design Manual supplement AISC 341?

Answer

The AISC Seismic Design Manual, 3rd edition, provides design examples, connection prequalification guidance, and practical application of AISC 341 provisions, helping engineers correctly implement ductility detailing requirements for each seismic force-resisting system category.

#197 · Steel Lateral Force-Resisting Systems

What is the design implication of a lower R factor for OCBF versus SCBF?

Answer

A lower R factor increases the design base shear the frame must resist elastically, reflecting OCBF's reduced ductility capacity, so an OCBF frame typically requires larger member sizes than an SCBF frame designed for the identical seismic demand.

#198 · Steel Lateral Force-Resisting Systems

Why are K-braced configurations prohibited in seismic steel bracing?

Answer

In a K-brace configuration the braces connect to a column at a point between floors, so brace buckling or yielding induces a large unbalanced lateral force on the column, potentially causing column failure, which is why AISC 341 prohibits this configuration in SCBF and OCBF.

#199 · Steel Lateral Force-Resisting Systems

What is the expected brace strength used for in SCBF capacity design?

Answer

Expected brace strength, computed using expected yield stress R_y and expected tensile strength R_t , is used to determine the maximum forces the brace can deliver to connections, gusset plates, beams, and columns, ensuring those elements remain elastic under the ductile brace's full capacity.

#200 · Steel Lateral Force-Resisting Systems

What is the role of link rotation angle limits in EBF design?

Answer

AISC 341 limits the inelastic rotation angle a link may undergo, typically 0.08 radians for shear links, ensuring the link can sustain the expected cyclic deformation demand from the design earthquake without excessive strength or stiffness degradation.

#201 · Steel Lateral Force-Resisting Systems

How is link length classified in EBF design?

Answer

Link length is classified as shear-critical, intermediate, or flexure-critical based on comparing the link length to the ratio of plastic moment capacity to plastic shear capacity of the link section, which governs whether shear or flexural yielding controls behavior.

#202 · Steel Lateral Force-Resisting Systems

What lateral bracing requirement applies to EBF link ends?

Answer

Both ends of an EBF link must be provided with lateral bracing to prevent lateral torsional buckling of the link during large inelastic rotations, since an unbraced link could twist out of plane and lose its energy dissipation capacity prematurely.

#203 · Steel Lateral Force-Resisting Systems

What is the purpose of stiffeners at EBF link ends and interior?

Answer

Web stiffeners at link ends and, for shorter shear links, along the link length control local web buckling and stabilize the yielding web panel, allowing the link to achieve its full inelastic rotation capacity without premature local instability.

#204 · Steel Lateral Force-Resisting Systems

What connection type is typically used for EBF brace-to-beam joints outside the link?

Answer

Brace connections outside the link are designed using capacity design forces based on the expected strength of the link, ensuring the connection and the brace itself remain essentially elastic while all inelastic action concentrates in the designated link.

#205 · Steel Lateral Force-Resisting Systems

Why is width-to-thickness ratio limited for ductile steel members?

Answer

AISC 341 imposes highly ductile or moderately ductile width-to-thickness limits on flanges and webs of members expected to undergo inelastic action, preventing local buckling from prematurely degrading strength before the member achieves its required plastic rotation capacity.

#206 · Steel Lateral Force-Resisting Systems

What is the significance of the response modification coefficient R for steel systems?

Answer

R accounts for a system's ductility and overstrength, reducing the elastic seismic force to a design level; systems with more reliable ductile detailing, such as SMF, receive a higher R than less ductile systems like OMF, lowering their required design strength.

#207 · Steel Lateral Force-Resisting Systems

What does the overstrength factor Ω_0 represent for steel lateral systems?

Answer

Ω_0 estimates the ratio between the maximum force a system can develop, considering material overstrength and strain hardening, and the design seismic force, and is used to amplify forces on elements requiring capacity design protection, such as columns below braces.

#208 · Steel Lateral Force-Resisting Systems

What column splice requirement applies in steel seismic force-resisting systems?

Answer

Column splices in the seismic force-resisting system must be designed for amplified seismic forces and located away from regions of high inelastic demand, with adequate flexural and shear strength to maintain load path continuity during a design earthquake.

#209 · Steel Lateral Force-Resisting Systems

What is a buckling restrained braced frame (BRBF) and how does it differ from SCBF?

Answer

A BRBF uses a steel core encased in a debonded restraining sleeve, preventing brace buckling entirely so the brace yields in both tension and compression symmetrically, providing more stable hysteretic behavior and higher ductility than a conventional SCBF brace.

#210 · Steel Lateral Force-Resisting Systems

Why must chevron (V or inverted-V) braced frame beams resist an unbalanced vertical force?

Answer

When one brace buckles in compression and the other yields in tension at a chevron connection point, the vertical force components no longer balance, so the beam must be designed for the resulting unbalanced load using the braces' expected strengths.

#211 · Steel Lateral Force-Resisting Systems

What is the purpose of the highly ductile member classification in AISC 341?

Answer

Highly ductile members must satisfy the most restrictive width-to-thickness ratios, ensuring they can sustain large inelastic rotations without local buckling, and are required for elements like SMF beams and EBF links that undergo the greatest cyclic inelastic demand.

#212 · Steel Lateral Force-Resisting Systems

How does connection prequalification affect SMF beam-column joint design?

Answer

AISC 358 prequalified connections, such as reduced beam section or bolted flange plate details, have been cyclically tested and satisfy AISC 341 criteria, allowing engineers to use them without project-specific testing, provided all prequalification limits on size and geometry are met.

#213 · Steel Lateral Force-Resisting Systems

What is the continuity plate requirement at SMF beam-column joints?

Answer

Continuity plates, welded to the column web and flanges, transfer beam flange forces through the panel zone and prevent local column flange bending, and are required unless the column flange is sufficiently thick to resist the concentrated beam flange force alone.

#214 · Steel Lateral Force-Resisting Systems

What is the significance of drift compatibility for steel gravity connections in high seismic zones?

Answer

Gravity-only connections not part of the seismic force-resisting system must accommodate the design story drift without loss of vertical load carrying capacity, since these connections still deform along with the frame during a design earthquake.

#215 · Steel Lateral Force-Resisting Systems

Why is a diagonal brace preferred over a K-brace for seismic force resistance?

Answer

A diagonal brace, connecting corner to corner of a bay, transfers axial force directly into the beam-column joint or into a beam that can be capacity designed, avoiding the unbalanced mid-height column force that a K-brace configuration would create.

#216 · Steel Lateral Force-Resisting Systems

What effect does brace slenderness ratio have on SCBF hysteretic behavior?

Answer

AISC 341 limits brace slenderness to control post-buckling strength degradation; a more slender brace loses compressive capacity more severely after buckling, so limiting slenderness helps maintain more stable, predictable energy dissipation over repeated seismic cycles.

#217 · Steel Lateral Force-Resisting Systems

What is the design intent behind requiring a balanced tension-compression brace pattern in SCBF?

Answer

AISC 341 requires that along any line of bracing, at least 30 percent but not more than 70 percent of the total horizontal force be resisted by braces in tension, preventing a system that relies entirely on compression braces from losing significant capacity after buckling.

#218 · Steel Lateral Force-Resisting Systems

What is the function of a collector beam in a steel braced frame system?

Answer

A collector beam gathers diaphragm shear forces from a wide tributary area and delivers them to the vertical bracing element, and must be designed for amplified seismic forces since it is a critical load path component whose failure would disrupt force transfer.

#219 · Steel Lateral Force-Resisting Systems

How does torsional restraint requirement apply to SCBF brace connections?

Answer

Brace connections must be designed to accommodate the brace's post-buckling flexural demand out of plane, since the brace end rotates as it buckles, requiring the gusset plate to permit this rotation without developing unintended moment restraint that could fracture the connection.

#220 · Steel Lateral Force-Resisting Systems

What is the difference in expected inelastic behavior between SMF and SCBF systems under a design earthquake?

Answer

SMF systems dissipate energy through distributed flexural yielding at beam ends over many stories, while SCBF systems concentrate energy dissipation in axial brace yielding and buckling, generally giving SMF a more distributed but more flexible drift response.

#221 · Steel Lateral Force-Resisting Systems

What is the design purpose of doubler plates in SMF panel zones?

Answer

Doubler plates increase the shear strength of the column web panel zone when the unreinforced web cannot resist the shear demand generated by beam flange forces, helping satisfy the AISC 341 panel zone shear strength requirement at the beam-column joint.

#222 · Steel Lateral Force-Resisting Systems

Determine why OCBF systems are often selected for low-rise industrial buildings in low seismic zones.

Answer

OCBF systems have simpler, less costly connection detailing than SCBF since they are not required to accommodate large inelastic brace buckling deformation, making them economical where the lower R value and correspondingly higher design force still result in reasonable member sizes.

#223 · Steel Lateral Force-Resisting Systems

What is the significance of the expected yield stress R_y in steel seismic capacity design?

Answer

R_y is the ratio of expected yield stress to specified minimum yield stress and accounts for the fact that actual steel yield strength typically exceeds the nominal value, ensuring capacity-protected elements are designed for the true expected strength of the ductile fuse.

#224 · Steel Lateral Force-Resisting Systems

How does the redundancy factor ρ interact with steel lateral system design?

Answer

The redundancy factor ρ , per ASCE 7-16 Section 12.3.4, increases seismic design forces for structures in higher Seismic Design Categories that lack sufficient redundancy in their lateral system, such as too few bays of bracing or moment frame bays.

#225 · Steel Lateral Force-Resisting Systems

What is the significance of column base connection design in steel moment frames?

Answer

Column base connections in SMF and IMF systems must be designed to develop either the expected column strength or the maximum force the frame can deliver, since the base is a critical load path location where premature failure would compromise the entire lateral system.

#226 · Steel Lateral Force-Resisting Systems

Select the correct statement about the flexural link classification in an EBF.

Answer

A flexure-critical link is longer relative to its plastic moment-to-shear ratio and is expected to yield primarily in flexure at its ends, developing plastic hinges there rather than the shear yielding mechanism that governs shorter shear-critical links.

#227 · Steel Lateral Force-Resisting Systems

What is the role of the AISC 341 prequalification requirement for bolted connections in SMF?

Answer

Bolted moment connections used in SMF must be prequalified per AISC 358 or qualified by project-specific cyclic testing, ensuring the bolt group, end plate, or flange plate detail can reliably achieve the required plastic rotation without brittle bolt or weld fracture.

#228 · Steel Lateral Force-Resisting Systems

Calculate why a taller SCBF brace bay generally requires a heavier brace section than a shorter bay for the same story shear.

Answer

A taller bay produces a steeper brace angle and longer brace length, increasing the axial force needed to resist the same horizontal story shear and reducing the brace's buckling capacity due to increased slenderness, both of which drive a larger required brace section.

#229 · Steel Lateral Force-Resisting Systems

What is the purpose of limiting the number of link lengths used within a single EBF bay?

Answer

Using a consistent link length within a bay simplifies capacity design of braces, beams outside the link, and columns, since the expected link strength used to compute those protected forces depends directly on the specific link length and its shear-flexure classification.

#230 · Steel Lateral Force-Resisting Systems

Why does AISC 341 require special inspection for demand critical welds?

Answer

Demand critical welds occur at locations of the highest inelastic strain demand in SMF and EBF systems, so special inspection, including ultrasonic testing, verifies weld quality and the absence of flaws that could initiate brittle fracture during a design earthquake.

#231 · Concrete Lateral Force-Resisting Systems

What ACI 318-14 chapter governs special seismic detailing for concrete structures?

Answer

ACI 318-14 Chapter 18 governs earthquake-resistant structures, providing detailing requirements for special, intermediate, and ordinary moment frames, but for cast-in-place shear walls only a special and ordinary tier exist (no intermediate tier for CIP shear walls), with the tier required depending on the assigned Seismic Design Category.

#232 · Concrete Lateral Force-Resisting Systems

What is a special reinforced concrete shear wall (SRCSW)?

Answer

A special reinforced concrete shear wall satisfies ACI 318-14 Chapter 18 ductile detailing requirements, including boundary element confinement and minimum reinforcement ratios, and is required in higher Seismic Design Categories to provide reliable inelastic deformation capacity.

#233 · Concrete Lateral Force-Resisting Systems

What is the purpose of a boundary element in a concrete shear wall?

Answer

A boundary element is a confined region at the wall edge, detailed with closely spaced transverse reinforcement, that resists concentrated compressive strains and prevents concrete crushing and bar buckling as the wall undergoes flexural yielding under seismic loading.

#234 · Concrete Lateral Force-Resisting Systems

How does ACI 318-14 determine whether special boundary elements are required in a shear wall?

Answer

ACI 318-14 Section 18.10.6 provides two methods, a displacement-based approach checking the neutral axis depth against wall length and design displacement, and a stress-based approach checking extreme fiber compressive stress against a specified threshold under factored loads.

#235 · Concrete Lateral Force-Resisting Systems

What is a coupling beam in a concrete shear wall system?

Answer

A coupling beam links two wall piers across an opening, such as a doorway, transferring shear and moment between the piers and engaging both walls to act partially as a single composite system, which increases overall lateral stiffness and energy dissipation.

#236 · Concrete Lateral Force-Resisting Systems

Why do some coupling beams use diagonal reinforcement instead of conventional stirrups?

Answer

Diagonally reinforced coupling beams, required by ACI 318-14 for beams with high shear demand relative to span, provide superior ductility and resistance to sliding shear failure along the beam depth compared to conventionally reinforced beams under large cyclic seismic loading.

#237 · Concrete Lateral Force-Resisting Systems

What distinguishes a special moment frame (SMF) from an intermediate moment frame (IMF) in concrete?

Answer

A special concrete moment frame requires strict strong column weak beam design, confined plastic hinge regions with tight hoop spacing, and joint shear checks per ACI 318-14 Chapter 18, while an intermediate moment frame has relaxed detailing suited to lower seismic demand.

#238 · Concrete Lateral Force-Resisting Systems

What is the strong column weak beam requirement in ACI 318-14 special moment frames?

Answer

The sum of nominal column flexural strengths at a joint must be at least 1.2 times the sum of nominal beam flexural strengths framing into that joint, forcing plastic hinges to form in beams rather than columns to preserve overall frame stability.

#239 · Concrete Lateral Force-Resisting Systems

What is the purpose of transverse hoop reinforcement in special moment frame beam plastic hinge regions?

Answer

Closely spaced hoops in the plastic hinge region confine the concrete core, restrain longitudinal bar buckling, and provide shear strength during large inelastic flexural cycles, allowing the beam to sustain the required plastic rotation without brittle shear failure.

#240 · Concrete Lateral Force-Resisting Systems

What is joint shear design intended to prevent in special moment frames?

Answer

Joint shear design ensures the beam-column joint core can transfer the large shear force generated by opposing beam flexural yielding without shear failure, since joint failure would disconnect the beam from the column and compromise the entire load path.

#241 · Concrete Lateral Force-Resisting Systems

What is a diaphragm collector in concrete floor system design?

Answer

A collector is a reinforced concrete or embedded steel element that gathers diaphragm shear from a distributed slab region and delivers it as a concentrated force to a vertical lateral element such as a shear wall, requiring design for amplified seismic collector forces.

#242 · Concrete Lateral Force-Resisting Systems

How is diaphragm shear transferred at the connection between a slab and a shear wall?

Answer

Diaphragm shear transfers through shear friction reinforcement, dowels, or direct bearing at the slab-to-wall interface, with the connection designed for the calculated diaphragm shear demand amplified as required for the assigned Seismic Design Category.

#243 · Concrete Lateral Force-Resisting Systems

What is an ordinary reinforced concrete shear wall and where is it permitted?

Answer

An ordinary reinforced concrete shear wall follows only ACI 318-14 Chapter 11 provisions without Chapter 18 ductile detailing, and per ASCE 7-16 Table 12.2-1 is generally limited to lower Seismic Design Categories due to its lower R value and limited ductility.

#244 · Concrete Lateral Force-Resisting Systems

What minimum reinforcement requirement applies to special reinforced concrete shear walls?

Answer

ACI 318-14 requires minimum distributed horizontal and vertical reinforcement ratios in special shear walls, typically 0.0025, to control cracking and provide distributed ductility, with maximum bar spacing limits to ensure reinforcement engages the full wall section.

#245 · Concrete Lateral Force-Resisting Systems

What is the purpose of checking wall shear strength against the nominal flexural capacity in special shear walls?

Answer

Shear design forces are based on the wall's probable flexural strength rather than the design seismic force alone, ensuring the wall yields in flexure, a ductile mechanism, before it can fail in shear, which is a brittle and less desirable failure mode.

#246 · Concrete Lateral Force-Resisting Systems

Why does ACI 318-14 require confinement reinforcement at the base of special shear walls?

Answer

The base of a special shear wall is the location of maximum flexural demand and expected plastic hinge formation during a design earthquake, so confined boundary elements are required there to sustain large compressive strain without concrete crushing or bar buckling.

#247 · Concrete Lateral Force-Resisting Systems

What is the significance of wall aspect ratio (height to length) for shear wall behavior?

Answer

A wall with a high aspect ratio behaves more flexurally, developing ductile flexural yielding at its base, while a squat wall with a low aspect ratio tends to be shear-dominated, which is less ductile and requires careful shear strength verification.

#248 · Concrete Lateral Force-Resisting Systems

What is a perforated shear wall and what design challenge does it present?

Answer

A perforated shear wall contains multiple openings, such as windows or doors, creating individual wall piers and coupling beams; each pier must be checked individually for combined flexure and shear, and coupling beams must be detailed for the resulting deep beam shear demand.

#249 · Concrete Lateral Force-Resisting Systems

Determine why a concrete special moment frame requires closely spaced transverse reinforcement at column ends.

Answer

Column ends near beam-column joints are potential plastic hinge locations under load reversal, and closely spaced hoops confine the concrete core, provide shear strength, and prevent longitudinal bar buckling, allowing the column to sustain axial load through large seismic drift cycles.

#250 · Concrete Lateral Force-Resisting Systems

What is the role of a diaphragm chord in a concrete floor diaphragm?

Answer

A diaphragm chord resists the tension and compression couple generated as the diaphragm bends like a deep beam spanning between vertical lateral elements, typically provided by continuous reinforcement at the diaphragm edge designed for the calculated chord force.

#251 · Concrete Lateral Force-Resisting Systems

Calculate why a longer shear wall generally has greater overturning resistance than a shorter wall for the same base shear.

Answer

A longer wall provides a larger lever arm between its tension and compression boundary elements, reducing the axial force demand at each boundary for a given overturning moment, which lowers required boundary element reinforcement and confinement compared to a shorter wall.

#252 · Concrete Lateral Force-Resisting Systems

What is the purpose of splice location restrictions for longitudinal reinforcement in special moment frame columns?

Answer

ACI 318-14 restricts lap splices to the column mid-height region away from potential plastic hinge zones at column ends, preventing splice failure in a region of high inelastic curvature demand where bond deterioration could occur during cyclic loading.

#253 · Concrete Lateral Force-Resisting Systems

How is a special reinforced concrete shear wall protected against a brittle shear failure?

Answer

ACI 318-14 does not define a named Ω_v shear amplification factor. Instead, the wall is designed for the shear corresponding to its probable flexural strength (a capacity-design check), or the analysis shear amplified by the ASCE 7-16 system overstrength factor Ω_0 , so the wall cannot fail in shear before it can develop its intended flexural yielding mechanism.

#254 · Concrete Lateral Force-Resisting Systems

Why must slab-column connections in concrete buildings without moment frames still be checked for drift compatibility?

Answer

Even when slab-column connections are not part of the seismic force-resisting system, they must be shown to accommodate the design story drift and maintain punching shear capacity without failure, since gravity connections still deform with the building during an earthquake.

#255 · Concrete Lateral Force-Resisting Systems

What is the purpose of the effective flange width consideration in T-beam special moment frame design?

Answer

Effective flange width determines how much slab reinforcement contributes to the beam's negative moment flexural strength, which affects the strong column weak beam check, since underestimating the flange contribution could understate actual beam capacity and fail the required column-to-beam strength ratio.

#256 · Concrete Lateral Force-Resisting Systems

What load combination consideration governs coupling beam design in coupled shear wall systems?

Answer

Coupling beams must be designed for the combined gravity and seismic shear and moment demand generated as the two wall piers rotate relative to each other under lateral load, using probable moment strengths at each pier to determine coupling beam design forces.

#257 · Concrete Lateral Force-Resisting Systems

Select the correct description of an intermediate moment frame's seismic hook detailing requirement.

Answer

Intermediate moment frames require hoops or stirrups with seismic hooks at member ends over a specified length to provide moderate confinement and shear capacity, though the spacing and confinement requirements are less stringent than those required for special moment frames.

#258 · Concrete Lateral Force-Resisting Systems

What is the significance of the axial load level on flexural ductility of a special moment frame column?

Answer

Higher axial compressive load reduces a concrete column's curvature ductility capacity, so ACI 318-14 imposes stricter detailing and confinement requirements as axial load increases, since high axial load combined with flexural yielding increases the risk of brittle compression failure.

#259 · Concrete Lateral Force-Resisting Systems

What is the design purpose of anchorage development length at diaphragm-to-wall connections?

Answer

Reinforcement transferring diaphragm shear or chord tension into a shear wall must be developed sufficiently on each side of the connection to ensure the bar can reach its design stress without pullout, maintaining load path continuity during seismic loading.

#260 · Concrete Lateral Force-Resisting Systems

Determine why a coupled shear wall system typically has greater lateral stiffness than two isolated wall piers of the same total length.

Answer

Coupling beams engage both wall piers to act partially as a single composite unit, transferring shear between piers and creating a stiffer overall system, since the piers resist a shared overturning moment more efficiently than each pier acting independently.

#261 · Concrete Lateral Force-Resisting Systems

What is the role of horizontal reinforcement in a special shear wall web?

Answer

Horizontal web reinforcement resists shear demand across the wall length and provides distributed crack control, with ACI 318-14 requiring minimum ratios and spacing limits to ensure the reinforcement effectively engages the wall panel under cyclic shear loading.

#262 · Concrete Lateral Force-Resisting Systems

Why does ACI 318-14 require checking sliding shear at construction joints in special shear walls?

Answer

Construction joints, particularly at the wall base, represent a plane of weaker shear transfer than monolithic concrete, so shear friction reinforcement crossing the joint must be verified adequate to resist sliding shear demand without excessive slip during a design earthquake.

#263 · Concrete Lateral Force-Resisting Systems

What is the effect of a discontinuous shear wall (transfer condition) on foundation design?

Answer

A shear wall that does not extend to the foundation, requiring a transfer beam or diaphragm, concentrates large forces at the transfer level, requiring amplified design forces for the transfer element and careful load path verification to the foundation below.

#264 · Concrete Lateral Force-Resisting Systems

What is the purpose of checking column shear strength against probable moment capacity in special moment frames?

Answer

Column shear demand is computed using the probable flexural strengths of the beams framing into the column, ensuring the column has adequate shear capacity to remain elastic while the beams yield in flexure, avoiding a brittle column shear failure mechanism.

#265 · Concrete Lateral Force-Resisting Systems

Calculate why increasing boundary element confinement length reduces the risk of compression failure at a shear wall edge.

Answer

A longer confined boundary region distributes the concentrated compressive strain over a greater portion of the wall cross section, reducing peak strain demand at any single location and providing more reliable ductile deformation capacity before concrete crushing occurs.

#266 · Concrete Lateral Force-Resisting Systems

What is the difference between a special moment frame and a special shear wall in terms of primary lateral resistance mechanism?

Answer

A special moment frame resists lateral load through beam and column flexure with ductile hinging distributed over building height, while a special shear wall resists lateral load primarily through in-plane shear and flexural action concentrated at the wall base.

#267 · Concrete Lateral Force-Resisting Systems

What is the significance of diaphragm design forces per ASCE 7-16 for concrete floor systems in higher Seismic Design Categories?

Answer

ASCE 7-16 requires diaphragm design forces that may exceed the story force used for the vertical lateral system, since diaphragms must transfer forces reliably to all vertical elements, and inadequate diaphragm strength could prevent proper force distribution during an earthquake.

#268 · Concrete Lateral Force-Resisting Systems

What is the purpose of a drag strut in a concrete diaphragm with a shear wall not spanning the full building width?

Answer

A drag strut collects diaphragm shear from the portion of floor beyond the wall's own width and drags it into the shear wall, ensuring the wall receives the full tributary lateral force even though it does not extend across the entire diaphragm.

#269 · Concrete Lateral Force-Resisting Systems

Why is P-delta effect particularly important to check in slender special shear wall buildings?

Answer

Slender shear wall buildings can experience significant secondary moment from gravity loads acting through lateral displacement, which can amplify wall flexural demand and drift beyond first-order analysis results, requiring explicit P-delta consideration per ASCE 7-16.

#270 · Concrete Lateral Force-Resisting Systems

What load path consideration applies to out-of-plane wall forces in a concrete shear wall building?

Answer

Walls must be designed for out-of-plane seismic forces per ASCE 7-16, in addition to their in-plane shear wall function, since a wall segment can be subjected to out-of-plane demand from its own inertial mass independent of the diaphragm shear it resists in-plane.

#271 · Concrete Lateral Force-Resisting Systems

Determine why a special moment frame column must satisfy both flexural and shear capacity design checks simultaneously.

Answer

A column must have flexural strength satisfying the strong column weak beam ratio while also having shear strength based on probable beam moments, since satisfying only one check could still allow a brittle shear failure or a weak column flexural mechanism to govern.

#272 · Concrete Lateral Force-Resisting Systems

What is the purpose of the maximum longitudinal reinforcement ratio limit in special moment frame beams?

Answer

Limiting the maximum reinforcement ratio in special moment frame beams ensures the section behaves in a tension-controlled, ductile manner at flexural yielding, avoiding an over-reinforced condition that could cause a brittle concrete compression failure before steel yielding occurs.

#273 • Concrete Lateral Force-Resisting Systems

What is the design significance of a concrete shear wall's neutral axis depth check for boundary elements?

Answer

The displacement-based method compares the calculated neutral axis depth under design displacement to a code-specified limit related to wall length; exceeding that limit indicates high compressive strain demand requiring special confined boundary elements at that wall edge.

#274 • Concrete Lateral Force-Resisting Systems

Select the correct reason ACI 318-14 requires closely spaced hoops through beam-column joints in special moment frames.

Answer

Hoops through the joint region provide confinement to the joint core concrete and help resist the high joint shear stress generated when beams on both sides of the column develop plastic hinges simultaneously under seismic load reversal.

#275 · Concrete Lateral Force-Resisting Systems

What is the role of shear friction design in concrete diaphragm-to-collector connections?

Answer

Shear friction reinforcement crossing the interface between a diaphragm slab and a collector element provides clamping force that generates frictional resistance, transferring diaphragm shear into the collector without relying solely on concrete tensile or bond strength.

#276 · Concrete Lateral Force-Resisting Systems

Why is punching shear a critical concern at slab-to-wall connections resisting concentrated seismic collector forces?

Answer

A concentrated collector force delivered through a thin slab into a wall or column can generate high shear stress in a small area, so punching shear capacity must be verified adequate to prevent local slab failure at the force transfer location.

#277 · Concrete Lateral Force-Resisting Systems

What is the significance of hoop spacing limits at the top and bottom of special moment frame columns?

Answer

Reduced hoop spacing over a specified length at column ends, where plastic hinging is most likely, provides adequate confinement and shear strength for the anticipated inelastic rotation demand while less critical mid-height regions may use wider hoop spacing.

#278 · Concrete Lateral Force-Resisting Systems

Calculate the design implication of a squat shear wall with an aspect ratio less than one under lateral load.

Answer

A squat wall with height less than its length behaves as a deep beam dominated by shear rather than flexure, requiring careful shear strength design since its ductility is limited compared to a slender, flexure-dominated wall of higher aspect ratio.

#279 · Concrete Lateral Force-Resisting Systems

What is the function of a slab collector reinforcement band adjacent to a shear wall opening?

Answer

A slab collector band concentrates reinforcement to gather diaphragm shear from the region interrupted by a floor opening near the wall and route it around the opening into the wall, maintaining diaphragm load path continuity despite the discontinuity.

#280 · Concrete Lateral Force-Resisting Systems

Determine why coupling beam stiffness assumptions significantly affect coupled shear wall analysis results.

Answer

Coupling beam stiffness governs how much moment and shear transfer between wall piers, so an overly stiff assumption overestimates coupling action and understates individual pier demand, while an overly flexible assumption could understate coupling beam design forces themselves.

#281 · Concrete Lateral Force-Resisting Systems

What is the purpose of transverse reinforcement in diagonally reinforced coupling beams?

Answer

Transverse reinforcement confines each diagonal reinforcement bundle within the coupling beam, preventing bar buckling under the large compressive strain reversals the diagonal bars experience as the beam cycles between positive and negative bending during a design earthquake.

#282 · Concrete Lateral Force-Resisting Systems

What is the effect of increasing shear wall thickness on out-of-plane stability during flexural yielding?

Answer

Increased wall thickness improves out-of-plane stability of the compression boundary region during large inelastic flexural cycles, reducing the risk of lateral instability of the boundary element, which is a documented failure mode in thin, heavily loaded special shear walls.

#283 · Concrete Lateral Force-Resisting Systems

Select the correct statement about diaphragm design force levels in structures assigned to higher Seismic Design Categories.

Answer

ASCE 7-16 requires diaphragm design forces based on either the minimum diaphragm force provisions or amplified forces derived from the vertical system's overstrength, whichever governs, recognizing that diaphragms must remain capable of redistributing load if a vertical element yields.

#284 · Concrete Lateral Force-Resisting Systems

What is the significance of continuous top and bottom reinforcement through a special moment frame beam-column joint?

Answer

Continuous longitudinal reinforcement through the joint core ensures adequate development and force transfer between the beam and column at the joint, preventing anchorage failure that could occur if bars were terminated within the highly stressed joint region.

#285 · Concrete Lateral Force-Resisting Systems

Determine why designers check both flexural and axial-flexural interaction for special shear wall boundary elements.

Answer

Boundary elements resist combined axial compression from overturning and gravity load along with the wall's overall flexural demand, so checking the interaction ensures the confined region has adequate strength for the actual combined stress state rather than either effect alone.

#286 · Concrete Lateral Force-Resisting Systems

What is the purpose of anchorage requirements for boundary element transverse reinforcement in special shear walls?

Answer

Transverse confinement reinforcement, such as hoops or crossties, at shear wall boundary elements must be properly anchored with seismic hooks engaging longitudinal bars, ensuring the confinement remains effective and does not open up under large cyclic compressive and tensile strain.

#287 · Concrete Lateral Force-Resisting Systems

What is the design significance of vertical reinforcement continuity across floor levels in special shear walls?

Answer

Vertical reinforcement must be spliced or continuous across floor levels to maintain flexural strength continuity up the wall height, since a discontinuity or inadequately spliced bar could create a weak plane that limits the wall's overall ductile flexural mechanism.

#288 · Concrete Lateral Force-Resisting Systems

Calculate why increasing coupling beam depth generally increases coupling action in a coupled wall system.

Answer

A deeper coupling beam has greater flexural stiffness and moment capacity for the same span, allowing it to transfer more moment between wall piers, which increases the coupling effect and shifts more overturning resistance into the axial tension-compression couple of the piers.

#289 · Concrete Lateral Force-Resisting Systems

What is the role of ACI 318-14 in specifying minimum shear wall thickness requirements?

Answer

ACI 318-14 specifies minimum wall thickness related to unsupported height or length, controlling out-of-plane buckling risk and ensuring adequate cover for reinforcement, particularly important at boundary elements carrying high compressive stress under seismic loading.

#290 · Concrete Lateral Force-Resisting Systems

Determine why a designer would choose a coupled shear wall system over two independent isolated walls for a building with door openings.

Answer

A coupled system uses the openings' coupling beams to engage both piers in resisting overturning moment together, improving lateral stiffness and reducing individual pier demand compared to two fully isolated walls that would each resist their own overturning moment independently.

#291 · Wood and Masonry Lateral Force-Resisting Systems

What standard governs special design provisions for wood shear walls and diaphragms?

Answer

SDPWS 2015, the Special Design Provisions for Wind and Seismic published alongside NDS 2018, provides nailing schedules, shear capacities, and aspect ratio limits for wood structural panel shear walls and diaphragms used as lateral force-resisting elements.

#292 · Wood and Masonry Lateral Force-Resisting Systems

What is the maximum height-to-width aspect ratio typically permitted for a blocked wood structural panel shear wall?

Answer

SDPWS 2015 generally limits blocked wood structural panel shear walls to a 2 to 1 height-to-width aspect ratio, beyond which the wall's shear capacity must be reduced by a penalty factor to account for reduced anchorage effectiveness and increased overturning demand.

#293 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of blocking in a wood structural panel shear wall?

Answer

Blocking provides a nailing surface at horizontal panel edges between studs, allowing edge nailing along all four sides of each panel, which significantly increases shear capacity compared to an unblocked wall that only nails along vertical stud edges.

#294 · Wood and Masonry Lateral Force-Resisting Systems

How does nail spacing at panel edges affect wood shear wall design capacity?

Answer

Closer nail spacing at panel edges increases the tabulated unit shear capacity of a wood structural panel shear wall in SDPWS 2015 tables, since more closely spaced fasteners share the shear load along the panel edge and increase the wall's overall strength.

#295 · Wood and Masonry Lateral Force-Resisting Systems

What is a hold-down device and why is it required at wood shear wall ends?

Answer

A hold-down device anchors the shear wall end post to the foundation or floor below, resisting the net uplift tension force generated by overturning moment, preventing the wall from rotating or lifting at its tension end under lateral load.

#296 · Wood and Masonry Lateral Force-Resisting Systems

What is the function of a diaphragm chord in a wood floor or roof diaphragm?

Answer

A diaphragm chord, typically the top plate or a designated framing member at the diaphragm edge, resists the tension and compression forces generated as the diaphragm acts like a horizontal deep beam spanning between vertical shear wall lines.

#297 · Wood and Masonry Lateral Force-Resisting Systems

What distinguishes a flexible diaphragm from a rigid diaphragm in wood construction?

Answer

A flexible diaphragm, such as a wood structural panel roof or floor, deflects significantly relative to the shear walls it connects and distributes load to walls by tributary area, whereas a rigid diaphragm distributes load in proportion to relative wall stiffness.

#298 · Wood and Masonry Lateral Force-Resisting Systems

What is the effect of using a thinner wood structural panel on shear wall capacity?

Answer

A thinner wood structural panel generally has lower tabulated shear capacity per SDPWS 2015 for a given nail size and spacing, since panel thickness affects the panel's resistance to nail bearing and shear buckling under racking load.

#299 · Wood and Masonry Lateral Force-Resisting Systems

What is a perforated shear wall in wood-frame construction?

Answer

A perforated shear wall is a continuous wall line containing door or window openings, designed using a method that reduces tabulated shear capacity based on the percentage of full-height sheathing present, without requiring individual hold-downs at every opening.

#300 · Wood and Masonry Lateral Force-Resisting Systems

How does the wood shear wall segmented method differ from the perforated shear wall method?

Answer

The segmented method treats each full-height sheathed wall pier between openings as an individual shear wall segment requiring its own hold-downs, while the perforated method treats the entire wall line as one unit with a capacity reduction factor and end anchorage only.

#301 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a drag strut in a wood diaphragm system?

Answer

A drag strut, often the top plate of a wall not aligned with the diaphragm shear transfer path, collects diaphragm shear from a tributary width and delivers it into a shear wall, ensuring the wall receives its full design lateral force.

#302 · Wood and Masonry Lateral Force-Resisting Systems

What is the effect of specific gravity of framing lumber on wood shear wall nail capacity?

Answer

SDPWS 2015 tabulated shear capacities assume a minimum specific gravity for framing members, and lower density lumber requires a capacity adjustment factor since nail withdrawal and lateral capacity depend on the wood's density at the fastener location.

#303 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of diaphragm span-to-depth ratio limits in wood diaphragm design?

Answer

SDPWS 2015 limits the span-to-depth ratio of wood diaphragms to control excessive deflection and ensure the diaphragm behaves reliably as a deep beam, since an overly long, narrow diaphragm could develop unacceptable drift or force distribution inaccuracies.

#304 · Wood and Masonry Lateral Force-Resisting Systems

What is a subdiaphragm and when is it used in wood construction?

Answer

A subdiaphragm is a smaller structural unit within a larger diaphragm used to transfer out-of-plane wall anchorage forces into the main diaphragm, particularly needed when the primary diaphragm framing does not run in a direction suitable for direct anchorage force transfer.

#305 · Wood and Masonry Lateral Force-Resisting Systems

Why is out-of-plane wall anchorage a critical design consideration for wood diaphragms attached to concrete or masonry walls?

Answer

Concrete and masonry walls are heavy relative to the diaphragm and can pull away from the diaphragm under out-of-plane seismic inertial force if anchorage is inadequate, so ASCE 7-16 requires specific anchorage force levels and continuous ties for these wall-to-diaphragm connections.

#306 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of continuous cross ties in wood diaphragms supporting concrete or masonry walls?

Answer

Continuous cross ties transfer out-of-plane wall anchorage forces completely across the diaphragm to anchorage on the opposite side, preventing the diaphragm framing alone, which may be weak in cross-grain bending, from being the sole path resisting wall pull-off forces.

#307 · Wood and Masonry Lateral Force-Resisting Systems

What is the design significance of overturning restraint at multi-story wood shear wall stacking?

Answer

When shear walls stack over multiple stories, the cumulative overturning tension force must be transferred continuously from the top wall down through hold-downs at each level to the foundation, requiring a continuous tension load path check at each floor connection.

#308 · Wood and Masonry Lateral Force-Resisting Systems

What is a reinforced masonry shear wall per TMS 402/602-16?

Answer

A reinforced masonry shear wall contains vertical and horizontal steel reinforcement grouted into cells, providing ductile flexural and shear behavior that allows the wall to be used with a higher response modification coefficient than an unreinforced masonry wall.

#309 · Wood and Masonry Lateral Force-Resisting Systems

What is the primary limitation of unreinforced masonry shear walls in seismic design?

Answer

Unreinforced masonry shear walls lack ductile reinforcement to resist tension and redistribute stress after cracking, resulting in brittle failure modes, so ASCE 7-16 severely restricts or prohibits their use as the seismic force-resisting system in higher Seismic Design Categories.

#310 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of vertical reinforcement spacing limits in reinforced masonry shear walls?

Answer

TMS 402/602-16 specifies maximum vertical reinforcement spacing to ensure adequate distributed flexural capacity and crack control across the wall length, particularly important in special reinforced masonry shear walls required in higher Seismic Design Categories.

#311 · Wood and Masonry Lateral Force-Resisting Systems

What is out-of-plane anchorage for masonry walls and why is it critical in ASCE 7-16?

Answer

Out-of-plane anchorage connects masonry walls to the diaphragm to resist inertial forces perpendicular to the wall plane, and ASCE 7-16 Section 12.11 specifies minimum anchorage force requirements since masonry wall failure out-of-plane has historically caused significant earthquake damage and collapse.

#312 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of grouting requirements in reinforced masonry shear wall construction?

Answer

Grout fills cells containing reinforcement, bonding the steel to the masonry units and creating a composite section capable of resisting flexure and shear, with TMS 402/602-16 specifying grouting requirements to ensure adequate consolidation and reinforcement bond.

#313 · Wood and Masonry Lateral Force-Resisting Systems

What is a special reinforced masonry shear wall and when is it required?

Answer

A special reinforced masonry shear wall satisfies enhanced TMS 402/602-16 detailing including minimum reinforcement ratios and specific bar spacing, and is required in higher Seismic Design Categories where more reliable ductile behavior is needed for the assigned seismic demand.

#314 · Wood and Masonry Lateral Force-Resisting Systems

What is the design purpose of bond beams in reinforced masonry wall construction?

Answer

Bond beams are horizontal courses of grouted, reinforced masonry units, typically at floor and roof levels, that provide horizontal reinforcement continuity, diaphragm chord capacity, and improved out-of-plane wall stability at critical connection elevations.

#315 · Wood and Masonry Lateral Force-Resisting Systems

How does masonry shear wall pier design address flexure-shear interaction between openings?

Answer

Masonry piers between window or door openings must be checked for combined flexural and shear demand, similar to concrete wall piers, since the reduced pier width concentrates seismic force and can create critical stress conditions not present in a solid wall panel.

#316 · Wood and Masonry Lateral Force-Resisting Systems

What is the effect of masonry unit compressive strength (f'_m) on shear wall design capacity?

Answer

A higher specified masonry compressive strength f'_m increases the wall's flexural and shear capacity in TMS 402/602-16 design equations, since f'_m directly affects allowable compressive stress and the concrete-like shear contribution term in the masonry shear strength calculation.

#317 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of masonry wall slenderness ratio in out-of-plane design?

Answer

A higher slenderness ratio, the ratio of unsupported wall height to thickness, increases susceptibility to buckling under combined axial and out-of-plane bending, so TMS 402/602-16 imposes slenderness-based capacity reduction and prescriptive limits for masonry walls.

#318 · Wood and Masonry Lateral Force-Resisting Systems

Determine why masonry veneer attachment ties must be designed independently of the structural backup wall's lateral system.

Answer

Masonry veneer is a nonstructural cladding element that must transfer its own out-of-plane inertial force to the structural backup wall through ties, and this connection force is separate from and in addition to the backup wall's own seismic force-resisting design.

#319 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a bond beam at the top of a masonry parapet?

Answer

A reinforced bond beam at the parapet top provides continuity and out-of-plane stiffness to the unsupported cantilevered portion of the wall above the roofline, which is otherwise particularly vulnerable to out-of-plane seismic failure due to its cantilever configuration.

#320 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of the response modification coefficient R difference between special and ordinary reinforced masonry shear walls?

Answer

Special reinforced masonry shear walls receive a higher R value than ordinary reinforced masonry shear walls in ASCE 7-16 Table 12.2-1, reflecting their enhanced ductile detailing, which reduces the design seismic force but requires the additional reinforcement and spacing requirements.

#321 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why a wood shear wall with a 4 to 1 aspect ratio has lower usable shear capacity per unit length than one with a 2 to 1 aspect ratio.

Answer

SDPWS 2015 caps the permitted aspect ratio at 3.5 to 1 absolute maximum, and applies a strength-reduction factor of $2w/h$ to the tabulated unit shear capacity for any wall segment exceeding a 2 to 1 aspect ratio: a taller, narrower wall segment develops higher overturning demand relative to its shear-resisting length, making anchorage and panel shear transfer less efficient at greater slenderness, up to the point that a segment beyond 3.5 to 1 is not a permitted shear wall at all.

#322 · Wood and Masonry Lateral Force-Resisting Systems

What is the design significance of using oriented strand board versus plywood for wood shear wall sheathing?

Answer

Both oriented strand board and plywood are recognized wood structural panels in SDPWS 2015 with tabulated shear values, and the design capacity depends on the specific panel grade, thickness, and nail schedule rather than the panel type alone, provided grade requirements are met.

#323 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a diaphragm boundary nailing schedule distinct from field nailing?

Answer

Boundary and edge zones of a wood diaphragm typically require closer nail spacing than the panel field because shear demand is highest near diaphragm boundaries and shear walls, so SDPWS 2015 nailing schedules specify tighter spacing in these higher-demand zones.

#324 · Wood and Masonry Lateral Force-Resisting Systems

Determine why continuous tie-down rods stacked through multiple stories are used in tall, narrow wood shear wall segments.

Answer

A continuous tie-down rod system transfers accumulated overturning tension force directly from the roof level down through each floor to the foundation without relying on individual floor framing to resist that force, maintaining a reliable continuous uplift load path.

#325 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of a shear wall's unit shear capacity being expressed in pounds per linear foot?

Answer

Wood shear wall capacity is expressed in plf because sheathing and nailing resist shear distributed along the wall length, so total wall design shear strength is obtained by multiplying the tabulated unit shear value by the wall's actual sheathed length.

#326 · Wood and Masonry Lateral Force-Resisting Systems

What is the effect of increasing nail diameter on wood structural panel shear wall capacity?

Answer

A larger nail diameter generally increases the tabulated unit shear capacity in SDPWS 2015 tables because a larger fastener has greater lateral bearing and shear resistance, though panel thickness must also be adequate to prevent excessive nail head pull-through.

#327 · Wood and Masonry Lateral Force-Resisting Systems

Select the correct reason a rigid masonry diaphragm distributes lateral load differently than a flexible wood diaphragm.

Answer

A rigid diaphragm distributes shear to vertical elements in proportion to their relative lateral stiffness, requiring torsional analysis if the center of rigidity does not align with the center of mass, unlike a flexible diaphragm which distributes load simply by tributary area.

#328 · Wood and Masonry Lateral Force-Resisting Systems

What is the design significance of anchor bolt spacing at the sill plate of a wood shear wall?

Answer

Anchor bolt spacing at the sill plate must be designed to transfer the wall's total design shear into the foundation, with SDPWS 2015 and prescriptive code tables specifying maximum spacing based on the wall's shear capacity and the sill plate's bearing and shear resistance.

#329 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of checking cross-grain bending in wood ledger connections at masonry or concrete walls?

Answer

A wood ledger bolted to a masonry or concrete wall and relied upon to transfer out-of-plane wall anchorage force is prohibited from resisting that force through cross-grain bending, since wood is weak across the grain, requiring an alternative direct tie system instead.

#330 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of masonry shear wall reinforcement development length at wall-to-foundation dowel connections?

Answer

Vertical reinforcement dowels connecting a masonry shear wall to its foundation must be developed with adequate embedment length in both the foundation and the wall to ensure the reinforcement can reach its design tensile stress without pullout under seismic overturning demand.

#331 · Wood and Masonry Lateral Force-Resisting Systems

Determine why gypsum wallboard is generally assigned lower seismic shear capacity than wood structural panel sheathing in SDPWS 2015.

Answer

Gypsum wallboard is a more brittle material with lower shear stiffness and strength degradation under cyclic loading compared to wood structural panels, so SDPWS 2015 assigns gypsum wallboard significantly lower tabulated shear values and restricts its seismic application.

#332 · Wood and Masonry Lateral Force-Resisting Systems

What is the effect of double sided sheathing on wood shear wall design capacity?

Answer

When wood structural panels are applied to both faces of a shear wall, SDPWS 2015 permits the shear capacities of each side to be combined, subject to specific limitations, which can significantly increase the wall's total design shear strength for the same wall length.

#333 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a strap tie at a wood shear wall corner or intersection?

Answer

A strap tie provides direct tension force transfer across a wall corner or intersection where framing discontinuity would otherwise interrupt the diaphragm chord or collector load path, maintaining continuity of the tension force through the connection.

#334 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why increasing the number of shear wall lines in a wood building generally reduces the demand on any single wall line.

Answer

Adding shear wall lines increases the total available shear-resisting length distributing the story shear across more elements, which reduces the unit shear demand and overturning force each individual wall line must resist for the same total lateral force.

#335 · Wood and Masonry Lateral Force-Resisting Systems

What is the design significance of checking foundation sill plate bearing capacity in wood shear wall design?

Answer

The sill plate must resist compressive bearing stress from the shear wall's overturning compression toe force in addition to shear from the anchor bolts, requiring a bearing capacity check to prevent crushing of the sill plate under concentrated compressive load.

#336 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of TMS 402/602-16 prescriptive reinforcement requirements for masonry walls in Seismic Design Category D and higher?

Answer

TMS 402/602-16 mandates minimum vertical and horizontal reinforcement even in walls not explicitly designed as part of the lateral system in Seismic Design Category D and higher, ensuring baseline ductility and crack control throughout the structure regardless of specific design role.

#337 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a masonry infill wall within a steel or concrete frame requires special seismic consideration.

Answer

An infill wall not intended as part of the lateral system can still attract significant seismic force due to its stiffness relative to the surrounding frame, potentially altering the frame's expected behavior, so its interaction with the frame must be evaluated even if unintended.

#338 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of a wood diaphragm's unblocked versus blocked classification on allowable diaphragm span?

Answer

A blocked wood diaphragm, with edge nailing at all panel joints supported by blocking, achieves higher shear capacity and is generally permitted a longer allowable span-to-depth ratio than an unblocked diaphragm, which lacks that continuous edge support.

#339 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of checking net section capacity at a hold-down bolt location in a wood shear wall end post?

Answer

The bolt holes drilled for a hold-down device reduce the net cross-sectional area of the wood end post, so the post must be checked for adequate net section tension capacity to resist the overturning tension force without splitting or bolt hole tear-out failure.

#340 · Wood and Masonry Lateral Force-Resisting Systems

Select the correct description of how masonry shear wall shear strength is calculated per TMS 402/602-16.

Answer

Masonry shear wall nominal shear strength combines a masonry contribution term, based on cross-sectional area and specified compressive strength, and a steel contribution term from horizontal shear reinforcement, similar in concept to reinforced concrete shear strength calculation methods.

#341 · Wood and Masonry Lateral Force-Resisting Systems

What is the effect of grout type, fine versus coarse, on masonry wall construction per TMS 402/602-16?

Answer

Fine grout is used for smaller cell dimensions or narrower cavities while coarse grout is used for larger cells, with each type having different maximum pour height and consolidation requirements, ensuring adequate flow and consolidation around reinforcement without segregation.

#342 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a masonry shear wall pier's h/t ratio (height to thickness) matters for its axial load capacity.

Answer

A higher height-to-thickness ratio increases slenderness effects, reducing axial load capacity due to greater susceptibility to buckling under combined axial and any incidental lateral load, requiring TMS 402/602-16 slenderness reduction factors in the design calculation.

#343 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a masonry wall control joint and how does it interact with shear wall design?

Answer

A control joint accommodates shrinkage and thermal movement in masonry construction, and shear wall segments interrupted by control joints must be evaluated as separate structural elements, since the joint typically cannot transfer shear or flexural continuity across it.

#344 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why increasing wood shear wall sheathing thickness beyond the minimum code requirement rarely governs wall design.

Answer

Shear wall capacity is typically governed by the nailing schedule and fastener capacity rather than sheathing thickness alone once minimum panel thickness requirements are met, so increasing thickness without also tightening nail spacing usually does not increase the tabulated design shear value.

#345 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of continuous load path requirements connecting roof diaphragm, wall, and foundation in light-frame wood construction?

Answer

ASCE 7-16 and IBC 2018 require a continuous, positively connected load path from the roof diaphragm through each shear wall level down to the foundation, since any discontinuity in this chain would prevent seismic or wind forces from reaching the ground reliably.

#346 · Wood and Masonry Lateral Force-Resisting Systems

What is the design purpose of shear transfer connections between a wood diaphragm and a concrete tilt-up wall panel?

Answer

Shear transfer connections, such as embedded steel angles or continuous ledgers with mechanical fasteners, deliver diaphragm shear into the tilt-up wall panel, and must be designed for the calculated in-plane shear demand while the ledger itself avoids cross-grain bending for out-of-plane forces.

#347 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a wood shear wall with openings requires individual pier analysis under the segmented design method.

Answer

Each full-height sheathed segment between openings resists shear and overturning independently under the segmented method, so each pier's shear capacity, aspect ratio, and hold-down requirement must be checked separately rather than treating the wall line as a single continuous element.

#348 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of a masonry wall's reinforcement ratio limit to prevent brittle compression failure?

Answer

TMS 402/602-16 limits the maximum reinforcement ratio in flexural masonry design to ensure the section remains tension-controlled at the design flexural strength, preventing an over-reinforced brittle compression failure mode before the reinforcing steel yields.

#349 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a horizontal joint reinforcement or bond beam spacing limit in reinforced masonry construction?

Answer

Regularly spaced horizontal reinforcement or bond beams control shrinkage cracking, provide distributed horizontal shear resistance, and improve overall wall ductility, with TMS 402/602-16 specifying maximum vertical spacing limits especially in special reinforced masonry shear walls.

#350 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why a masonry shear wall with distributed reinforcement generally provides more reliable ductile behavior than one with concentrated reinforcement only at the ends.

Answer

Distributed vertical and horizontal reinforcement engages the full wall cross section in resisting flexure and shear, spreading cracking and yielding over a larger area, which produces more gradual, ductile degradation compared to a wall relying only on end reinforcement.

#351 · Wood and Masonry Lateral Force-Resisting Systems

What is the effect of using a metal connector such as a hold-down with a higher rated capacity on wood shear wall design economy?

Answer

A higher capacity hold-down can resist greater overturning tension force at a single wall end post, potentially allowing a shorter, more heavily loaded shear wall segment instead of requiring additional wall length or multiple lower-capacity devices to resist the same demand.

#352 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of checking the overturning restraint force at the lowest story of a multi-story wood shear wall building?

Answer

The lowest story hold-down and anchorage system must resist the cumulative overturning tension from all stories above, making it the critical connection in the continuous tension load path and typically requiring the largest capacity anchorage device in the building.

#353 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of checking diaphragm deflection in flexible wood diaphragm design for torsional irregularity evaluation?

Answer

Diaphragm deflection affects how lateral force distributes to shear walls and whether the diaphragm can be idealized as rigid or flexible for analysis purposes, and ASCE 7-16 requires this classification since it changes whether torsional irregularity provisions and amplification apply.

#354 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a masonry veneer over wood frame construction must be checked for out-of-plane seismic force independent of the wood shear wall system behind it.

Answer

Masonry veneer, though nonstructural, has significant mass generating its own inertial seismic force that must be transferred through ties to the structural wood backup wall, a design demand entirely separate from the backup wall's own in-plane shear wall design.

#355 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of TMS 402/602-16 requiring a minimum amount of horizontal reinforcement at the top of masonry walls?

Answer

Horizontal reinforcement at the top of a masonry wall, typically within a bond beam, ties the wall to the diaphragm and provides continuity for chord and collector forces, preventing separation at the wall-to-roof or wall-to-floor connection during seismic loading.

#356 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why a wood shear wall segment with a hold-down at only one end while relying on dead load at the other end may be inadequate under seismic reversal.

Answer

Seismic loading reverses direction, so each end of the wall alternately experiences net uplift tension; relying on dead load alone at one end is inadequate because during the reversed loading direction, that same end may require a hold-down that dead load cannot substitute for.

#357 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of an SDPWS 2015 shear wall type designation distinguishing wood structural panel from other sheathing materials?

Answer

SDPWS 2015 provides separate shear capacity tables for different sheathing types, such as wood structural panel, particleboard, or gypsum, since each material has distinct strength and stiffness properties under racking load, requiring type-specific design values rather than a single universal table.

#358 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of checking bearing wall capacity for combined gravity and lateral out-of-plane loads in a masonry building?

Answer

A masonry bearing wall must resist gravity axial load simultaneously with out-of-plane bending from seismic or wind forces, requiring an interaction check per TMS 402/602-16 since the combined effect can govern wall thickness and reinforcement even if each load considered alone would not.

#359 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a wood diaphragm chord splice location requires careful design attention.

Answer

A chord splice interrupts the continuous tension or compression member resisting diaphragm bending moment, so the splice connection, typically a strap or plate, must be designed to develop the full chord force at that location to maintain diaphragm flexural continuity.

#360 · Wood and Masonry Lateral Force-Resisting Systems

What is the design significance of the difference between wind and seismic governing load combinations for wood and masonry lateral systems?

Answer

SDPWS 2015 and TMS 402/602-16 detailing requirements often depend on which load, wind or seismic, governs and the assigned Seismic Design Category, since seismic ductility detailing can exceed what strength alone under wind load combinations would otherwise require.

#361 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why increasing masonry wall reinforcement without increasing wall thickness has a limited effect on out-of-plane slenderness capacity.

Answer

Out-of-plane slenderness capacity depends heavily on wall thickness through the radius of gyration and moment of inertia, so while additional reinforcement increases flexural strength, it does not address the slenderness reduction factor, which is primarily governed by the height-to-thickness ratio.

#362 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a continuity tie at wood diaphragm re-entrant corners?

Answer

Re-entrant corners create stress concentrations in diaphragm behavior due to the discontinuous diaphragm geometry, so continuity ties or drag struts are provided to maintain force transfer around the corner and prevent local diaphragm tearing or excessive local stress.

#363 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of TMS 402/602-16 requiring inspection of reinforcement placement before grouting in special reinforced masonry shear walls?

Answer

Verifying reinforcement size, spacing, and position before grouting ensures the as-built wall matches the design assumptions for flexural and shear capacity, since reinforcement placement errors would be impossible to detect or correct after the grout has been placed and cured.

#364 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a taller wood shear wall story typically requires a higher capacity hold-down than a shorter story for the same applied shear.

Answer

A taller story increases the moment arm between the applied lateral force and the wall base, increasing the overturning moment for the same shear force, which directly increases the required uplift tension capacity at the hold-down location.

#365 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a bond beam reinforcement lap splice length requirement in masonry shear wall construction?

Answer

Lap splices in bond beam horizontal reinforcement must be sufficiently long to develop the bar's full tensile strength across the splice, ensuring the horizontal reinforcement functions continuously for chord and shear resistance rather than as two disconnected segments.

#366 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of SDPWS 2015 permitting different nailing at diaphragm boundaries adjacent to shear walls versus interior panel edges?

Answer

Diaphragm boundary nailing adjacent to shear walls resists the highest shear transfer demand at the wall-diaphragm interface, so SDPWS 2015 typically specifies the closest nail spacing there compared to interior continuous panel edges, which see comparatively lower shear stress.

#367 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why the required overturning tension force at a shear wall increases as the wall's aspect ratio (height to length) increases for a given applied shear.

Answer

A greater height-to-length aspect ratio increases the lever arm relative to the wall's resisting width, so for the same applied shear force, the overturning moment divided by the shorter wall length produces a larger required tension force at the wall's uplift end.

#368 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of checking anchor bolt edge distance in concrete or masonry when anchoring wood sill plates?

Answer

Adequate edge distance for anchor bolts embedded in concrete or masonry prevents concrete cone breakout or masonry edge splitting failure under the applied shear load, ensuring the anchor bolt can develop its full design capacity for transferring shear wall shear into the foundation.

#369 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of masonry shear wall design requiring a check of both nominal shear strength and a maximum shear strength limit?

Answer

TMS 402/602-16 imposes an upper limit on nominal shear strength regardless of reinforcement quantity, preventing designers from relying on excessive shear reinforcement beyond what the masonry compression zone can actually support without a brittle diagonal compression failure mode.

#370 · Wood and Masonry Lateral Force-Resisting Systems

Determine why continuous vertical reinforcement at masonry wall corners and intersections is particularly important for seismic performance.

Answer

Corners and intersections experience biaxial stress conditions and are critical for maintaining overall building box behavior, so continuous, properly anchored vertical reinforcement there ensures load transfer between intersecting walls and prevents separation during seismic loading.

#371 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a shear wall design check comparing unit shear capacity to unit shear demand along the full wall length?

Answer

This comparison verifies the wall's sheathing and nailing provide sufficient design shear strength per unit length to meet or exceed the calculated seismic or wind unit shear demand distributed along the wall, confirming the selected nailing schedule is adequate before finalizing hold-down design.

#372 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of checking diaphragm-to-shear-wall connection capacity separately from the diaphragm's own shear capacity?

Answer

The connection transferring diaphragm shear into the shear wall top plate is a distinct load path element from the diaphragm sheathing itself, so its fastening, such as nailing or clip connectors, must be independently verified adequate for the calculated transfer shear force.

#373 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why increasing the number of anchor bolts along a wood shear wall sill plate reduces the shear demand per bolt.

Answer

Adding anchor bolts increases the total number of fasteners sharing the wall's total design shear force, so the shear force each individual bolt must resist decreases proportionally, provided the bolts are assumed to share load equally along the sill plate length.

#374 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of checking masonry wall reinforcement for adequate cover per TMS 402/602-16?

Answer

Minimum cover requirements protect reinforcement from corrosion and ensure adequate bond development between the steel and surrounding grout, with TMS 402/602-16 specifying cover based on exposure condition and reinforcement location within the wall cross section.

#375 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of TMS 402/602-16 shear reinforcement spacing limits along the height of a special reinforced masonry shear wall?

Answer

Maximum horizontal reinforcement spacing limits ensure the shear reinforcement engages the full wall height uniformly, preventing wide spacing that could leave unreinforced regions vulnerable to diagonal shear cracking under cyclic seismic loading demand.

#376 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a wood shear wall's compression chord (opposite the hold-down end) also requires design attention despite not needing uplift restraint.

Answer

The compression chord must resist a compressive bearing force equal in magnitude to the tension side's uplift demand, requiring adequate bearing capacity of the end post and sill plate to transfer that compressive overturning force into the foundation without crushing.

#377 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of verifying masonry shear wall flexural reinforcement anchorage at wall openings?

Answer

Reinforcement interrupted or routed around an opening must be properly anchored on each side with adequate development length, ensuring the reinforcement can still develop its full design tensile stress despite the discontinuity caused by the opening.

#378 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why a wood shear wall relying solely on gypsum wallboard sheathing would typically require significantly more wall length than one using wood structural panels for the identical seismic demand.

Answer

Because gypsum wallboard has substantially lower tabulated unit shear capacity per SDPWS 2015 than wood structural panel sheathing, a much greater total sheathed wall length is needed to accumulate sufficient combined shear capacity to resist the same total design lateral force.

#379 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of checking net uplift versus dead load restoring moment before requiring a hold-down at a wood shear wall end?

Answer

If the wall's dead load restoring moment exceeds the seismic or wind overturning moment at that end, a mechanical hold-down device may not be required, but this check must be performed explicitly rather than assumed, since underestimating overturning demand could leave the end unrestrained.

#380 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of TMS 402/602-16 requiring a minimum masonry wall reinforcement ratio even for walls resisting relatively low seismic demand in Seismic Design Category D?

Answer

Minimum prescriptive reinforcement provides a baseline level of ductility and redundancy independent of calculated demand, recognizing that seismic performance depends on more than just satisfying a strength check, particularly given the historically brittle failure record of underreinforced masonry.

#381 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a wood diaphragm supporting a masonry shear wall building requires stiffer design consideration than one supporting only wood shear walls.

Answer

Masonry shear walls are typically much stiffer than the flexible wood diaphragm, amplifying the relative flexibility mismatch, which increases diaphragm deflection demand and requires careful anchorage and subdiaphragm design to properly transfer force between the two dissimilar systems.

#382 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of checking combined axial and shear interaction for masonry wall piers between openings under lateral load?

Answer

A masonry pier between openings carries both gravity axial load and lateral shear simultaneously, and TMS 402/602-16 requires an interaction check since high axial compression can reduce the pier's effective shear capacity compared to a pier under lower axial demand.

#383 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why increasing wood shear wall length while keeping wall height constant improves the wall's aspect ratio compliance.

Answer

Increasing wall length while holding height constant reduces the height-to-length aspect ratio, which can bring an initially noncompliant wall segment within SDPWS 2015's allowable aspect ratio limit, avoiding the shear capacity penalty applied to overly narrow, tall segments.

#384 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of a masonry shear wall's design shear strength reduction factor (ϕ) per TMS 402/602-16?

Answer

The strength reduction factor accounts for uncertainty in material properties, construction quality, and the consequences of a shear failure mode, applied to the calculated nominal shear strength to arrive at a conservative design shear strength for comparison against factored shear demand.

#385 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of checking wood shear wall sill plate cross-grain tension resulting from hold-down bolt tension force?

Answer

A hold-down anchor bolt loaded in tension can induce cross-grain tension stress in the sill plate if not properly detailed, and since wood is weak in tension perpendicular to grain, the connection detail must route the tension force to avoid relying on this weak wood property.

#386 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a wood roof diaphragm framed with trusses generally requires the same chord and collector design principles as one framed with conventional rafters.

Answer

Regardless of the specific roof framing method, the diaphragm still functions as a horizontal deep beam requiring chords to resist bending-induced tension and compression and collectors to gather shear into shear walls, since these force transfer mechanisms depend on diaphragm geometry, not framing type.

#387 · Wood and Masonry Lateral Force-Resisting Systems

What is the purpose of checking masonry wall out-of-plane deflection limits under combined gravity and lateral load per TMS 402/602-16?

Answer

Deflection limits control secondary P-delta moment amplification in slender masonry walls under combined axial and out-of-plane bending, ensuring the wall does not experience excessive deflection that could lead to instability or unacceptable serviceability performance.

#388 · Wood and Masonry Lateral Force-Resisting Systems

Calculate why a masonry shear wall pier with a low height-to-length ratio between openings behaves differently in shear than a similar wood-frame pier.

Answer

A squat masonry pier tends toward diagonal shear cracking as the dominant failure mode given masonry's material behavior, requiring TMS 402/602-16 shear strength verification, whereas a comparably squat wood-frame pier's behavior is governed instead by SDPWS 2015 tabulated shear values and aspect ratio penalties specific to wood sheathing.

#389 · Wood and Masonry Lateral Force-Resisting Systems

What is the significance of a continuous positive connection requirement between a wood diaphragm and a masonry or concrete wall beyond simple bearing?

Answer

IBC 2018 and ASCE 7-16 require an explicit mechanical connection, not reliance on friction or bearing alone, between a wood diaphragm and masonry or concrete wall, since bearing connections have historically failed to reliably transfer out-of-plane anchorage forces during seismic events.

#390 · Wood and Masonry Lateral Force-Resisting Systems

Determine why a special reinforced masonry shear wall generally allows taller building heights under ASCE 7-16 Table 12.2-1 than an ordinary reinforced masonry shear wall for the same Seismic Design Category.

Answer

Special reinforced masonry shear walls satisfy stricter minimum reinforcement and detailing requirements that provide greater ductility and reliable inelastic deformation capacity, which justifies the code permitting greater structural height limits compared to the less ductile ordinary reinforced masonry shear wall category.

#391 · Connections and Structural System Integration

What is a collector element in a lateral force-resisting system?

Answer

A collector, also called a drag strut, gathers diaphragm shear over a tributary length and transfers it into a vertical lateral element such as a shear wall or braced frame, ensuring a continuous load path.

#392 · Connections and Structural System Integration

Why must collector connections be designed for amplified seismic loads?

Answer

ASCE 7-16 Section 12.10.2.1 requires collectors and their connections in structures assigned to SDC C through F to resist forces including the overstrength factor Ω_0 , since collector failure is brittle and non-ductile.

#393 · Connections and Structural System Integration

Define a diaphragm chord and its primary function.

Answer

A chord is a boundary member at the edge of a diaphragm that resists the tension and compression couple created by diaphragm bending, acting like the flange of a horizontal beam spanning between vertical supports.

#394 · Connections and Structural System Integration

What connection provides diaphragm-to-wall anchorage against out-of-plane forces?

Answer

Positive anchorage using steel straps, bolts, or embedded anchors ties the diaphragm to the wall, transferring out-of-plane wall forces into the diaphragm so the wall does not separate from the roof or floor during shaking.

#395 · Connections and Structural System Integration

Why does ASCE 7-16 restrict wood ledgers from resisting cross-grain bending in anchorage?

Answer

Cross-grain bending or tension in wood ledgers is a known brittle failure mode observed in earthquakes, so Section 12.11.2.2.3 requires anchorage forces to bypass the ledger through separate steel elements rather than loading it perpendicular to grain.

#396 · Connections and Structural System Integration

What is a continuity tie and why is it required across a building?

Answer

A continuity tie is a structural member interconnecting building elements so the structure acts as a unit; ASCE 7-16 Section 12.1.3 requires ties capable of transmitting seismic design forces induced by parts of the structure to the seismic force resisting system, with a minimum force not less than 5 percent of the weight of the smaller connected portion.

#397 · Connections and Structural System Integration

Describe the function of a brace gusset plate in a concentrically braced frame.

Answer

A gusset plate connects the diagonal brace to the beam and column, transferring axial brace forces into the frame while accommodating the brace end rotation demand that occurs during buckling and plastic hinge formation.

#398 · Connections and Structural System Integration

What is the purpose of a moment connection in a special moment frame?

Answer

A moment connection rigidly joins beam to column so that lateral loads produce beam bending and shear transfer at the joint, allowing the frame to develop ductile flexural hinging away from the column face under seismic demand.

#399 · Connections and Structural System Integration

Why must the load path be traceable from roof to foundation in every building?

Answer

Lateral forces generated at each level must have a complete, continuous path of elements and connections, from diaphragms through collectors and vertical elements to the foundation, or the structure cannot reliably resist wind or seismic demand.

#400 · Connections and Structural System Integration

What design force applies to connections of discontinuous shear wall systems?

Answer

ASCE 7-16 Section 12.3.3.3 requires elements supporting discontinuous walls or frames to be designed for the overstrength seismic load Ω_0 times the horizontal design force, since a sudden stiffness discontinuity concentrates demand.

#401 · Connections and Structural System Integration

What is a subdiaphragm and when is it used?

Answer

A subdiaphragm is a portion of the main diaphragm designed to transfer out-of-plane wall anchorage forces to nearby continuity ties or crossties, used when the main diaphragm chord or ties are not located directly at the wall line.

#402 · Connections and Structural System Integration

How does a boundary member differ from a general diaphragm chord?

Answer

A boundary member is any element at a diaphragm or shear wall edge, including openings, that resists axial forces from panel shear; a chord is specifically the perimeter boundary member resisting overall diaphragm bending forces.

#403 · Connections and Structural System Integration

What connection detail addresses shrinkage-compensating anchorage in wood ledger systems?

Answer

Steel anchor rods with washers, sleeves, or standoffs are used to bypass the ledger and connect directly to blocking or the diaphragm sheathing, avoiding reliance on the ledger's cross-grain capacity as wood shrinks over time.

#404 · Connections and Structural System Integration

Why are eccentrically braced frame link connections detailed differently than typical brace connections?

Answer

The link in an EBF is intended to yield in shear or flexure as the ductile fuse, so its connections and adjacent brace and beam connections must be designed to remain elastic while forcing inelastic action into the link itself.

#405 · Connections and Structural System Integration

What is the role of a positive connection at the top of a concrete shear wall to the roof diaphragm?

Answer

It transfers diaphragm shear and out-of-plane anchorage forces into the wall and provides continuity so the wall functions as intended vertical lateral support rather than relying on friction or gravity alone.

#406 · Connections and Structural System Integration

Define redundancy factor ρ and its effect on connection design forces in high seismic zones.

Answer

ρ accounts for the reliability of the lateral system based on element removal criteria; a ρ of 1.3 in SDC D through F increases seismic design forces by 30 percent on the horizontal seismic load, affecting connection demand.

#407 · Connections and Structural System Integration

What must a collector splice connection be designed to transfer?

Answer

A collector splice must transfer the full accumulated axial tension and compression force in the collector at that location, including the applicable overstrength factor, without relying on bearing alone across the joint.

#408 · Connections and Structural System Integration

How does diaphragm flexibility affect the distribution of collector forces?

Answer

In a flexible diaphragm, tributary areas are distributed by simple beam analogy to each vertical element, concentrating predictable collector demand at specific bays, while a rigid diaphragm distributes forces by relative stiffness across all elements.

#409 · Connections and Structural System Integration

What is a kicker or strut used for at a discontinuous collector line?

Answer

A kicker or strut member redirects collector force around an opening or offset in the diaphragm, maintaining a continuous load path when the collector cannot run in a straight line to the vertical element.

#410 · Connections and Structural System Integration

Why must anchor bolts connecting a steel base plate to concrete resist both shear and uplift?

Answer

Lateral load combinations can produce net uplift at column bases from overturning moment plus horizontal shear at the base, so anchor bolts and embedment must be sized for the combined tension and shear demand, not shear alone.

#411 · Connections and Structural System Integration

What is the purpose of a diaphragm-to-collector weld or bolt group design check?

Answer

It verifies that fastener spacing and capacity along the collector element can transfer the calculated shear flow from the diaphragm sheathing or deck into the collector without exceeding fastener or edge distance limits.

#412 · Connections and Structural System Integration

How does capacity design philosophy apply to brace connections in an SCBF?

Answer

Connections must be designed to develop the expected tensile yield and compressive buckling strength of the brace, including strain hardening, so the connection does not fail before the brace itself dissipates energy through ductile buckling.

#413 · Connections and Structural System Integration

What is meant by a strong column weak beam requirement in moment frame connection design?

Answer

AISC 341 requires the sum of column flexural strengths at a joint to exceed the sum of beam flexural strengths by a specified ratio, forcing plastic hinges to form in beams rather than columns to prevent story mechanism collapse.

#414 · Connections and Structural System Integration

Why is a positive mechanical connection required between precast diaphragm panels?

Answer

Precast panel joints lack the inherent continuity of cast-in-place concrete, so mechanical connectors or reinforced topping slabs are needed to transfer shear and maintain diaphragm continuity across panel boundaries.

#415 · Connections and Structural System Integration

What load combination governs uplift design at a shear wall hold-down connection?

Answer

The combination including $0.9D$ minus the seismic or wind overturning effect typically governs, since reduced dead load combined with full lateral overturning produces the maximum net uplift demand on the hold-down.

#416 · Connections and Structural System Integration

What is the term drag strut as distinct from a chord member?

Answer

A drag strut carries diaphragm shear parallel to its own axis toward a vertical resisting element, while a chord resists tension and compression perpendicular to the span direction created by diaphragm flexure; both may occur on the same member.

#417 · Connections and Structural System Integration

What connection detail prevents diaphragm punching or cross-grain splitting at hold-down bolts in wood construction?

Answer

Adequate edge distance, bolt spacing per NDS 2018, and steel bearing plates distribute the hold-down tension force over the wood cross-section, reducing local bearing stress and preventing splitting along the grain.

#418 · Connections and Structural System Integration

Why is continuity required between the roof diaphragm and parapet bracing elements?

Answer

Parapets are unsupported cantilevers subject to out-of-plane seismic force per ASCE 7-16 Chapter 13, and braces transferring that force into the diaphragm must connect continuously so the parapet does not overturn independently.

#419 · Connections and Structural System Integration

What is the design intent behind requiring positive connections at column splices in braced frames?

Answer

Column splices must maintain load path continuity for both gravity and lateral demand, including net tension that can develop from overturning in tall or slender braced bays, preventing separation under combined loading.

#420 · Connections and Structural System Integration

How does load path design differ between a rigid diaphragm building and a flexible diaphragm building at collector locations?

Answer

A rigid diaphragm shares collector demand proportionally with relative wall stiffness, potentially reducing peak collector force at any one line, while a flexible diaphragm assigns full tributary shear to the nearest collector independent of wall stiffness.

#421 · Foundations and Retaining Structures for Lateral Loads

What three failure modes must a spread footing under combined lateral and vertical load be checked against?

Answer

The footing must be checked for overturning about its toe, sliding along its base, and bearing capacity exceedance under the resultant eccentric vertical load combined with the applied lateral shear and moment.

#422 · Foundations and Retaining Structures for Lateral Loads

What factor of safety is typically required against overturning for a footing resisting lateral load?

Answer

A minimum factor of safety of 1.5 against overturning is commonly required, computed as the ratio of resisting moment from gravity loads to the overturning moment from applied lateral forces about the footing toe.

#423 · Foundations and Retaining Structures for Lateral Loads

What is passive pressure and how does it contribute to sliding resistance?

Answer

Passive pressure is the lateral soil resistance mobilized against the face of an embedded footing or wall as it tends to move into the soil, adding to base friction to resist the total applied sliding force.

#424 · Foundations and Retaining Structures for Lateral Loads

Compare Rankine and Coulomb lateral earth pressure theories.

Answer

Rankine theory assumes a frictionless wall-soil interface and a vertical wall face for simpler active and passive pressure coefficients, while Coulomb theory accounts for wall friction and a sloped wall face, giving more general but more complex results.

#425 · Foundations and Retaining Structures for Lateral Loads

What is the at-rest earth pressure coefficient K_0 used for?

Answer

K_0 represents lateral earth pressure on a wall that does not move or rotate, such as a basement wall braced top and bottom, and is typically estimated as 1 minus the sine of the soil friction angle for normally consolidated soil.

#426 · Foundations and Retaining Structures for Lateral Loads

Why does bearing pressure distribution become triangular or partial under high overturning moment?

Answer

When the resultant force falls outside the middle third of the footing base, tension would develop under rigid body assumptions, so the compressive bearing pressure redistributes over a reduced effective footing area to maintain equilibrium.

#427 · Foundations and Retaining Structures for Lateral Loads

What is meant by the middle third rule for footing eccentricity?

Answer

If the resultant vertical load eccentricity falls within the middle third of the footing width, the entire base remains in compression with a trapezoidal pressure distribution and no uplift or gapping occurs at the footing edge.

#428 · Foundations and Retaining Structures for Lateral Loads

How is the effective footing width calculated for a highly eccentric load per Meyerhof's method?

Answer

The effective width equals the actual footing width minus twice the eccentricity, centering a smaller uniform-pressure footing on the load resultant, which is then used to compute a reduced bearing capacity for design.

#429 · Foundations and Retaining Structures for Lateral Loads

What causes an increase in allowable bearing pressure under short-duration lateral loads such as wind or seismic?

Answer

Building codes commonly permit a one-third increase in allowable soil bearing pressure for load combinations that include transient wind or seismic effects, reflecting the short duration and low probability of sustained peak loading.

#430 · Foundations and Retaining Structures for Lateral Loads

How does base friction resist sliding at a spread footing?

Answer

Base friction resistance equals the normal force from vertical load at the footing base multiplied by the coefficient of friction between concrete and soil, providing the primary resistance against horizontal sliding demand.

#431 · Foundations and Retaining Structures for Lateral Loads

What is a key or shear key at the base of a retaining wall footing used for?

Answer

A shear key is a vertical projection extending below the footing base into undisturbed soil that mobilizes additional passive resistance, increasing total sliding resistance beyond base friction and adjacent passive pressure alone.

#432 · Foundations and Retaining Structures for Lateral Loads

Define active earth pressure and the condition required to develop it.

Answer

Active earth pressure is the minimum lateral pressure exerted by retained soil, developed when the wall moves away from the backfill enough to fully mobilize shear strength within the soil wedge behind the wall.

#433 · Foundations and Retaining Structures for Lateral Loads

How does surcharge loading behind a retaining wall affect the lateral earth pressure diagram?

Answer

A uniform surcharge behind the wall adds a constant lateral pressure increment over the full wall height, equal to the surcharge magnitude times the active earth pressure coefficient, shifting the pressure diagram upward uniformly.

#434 · Foundations and Retaining Structures for Lateral Loads

What is the design approach for a laterally loaded pile using the p-y method conceptually?

Answer

The p-y method models soil resistance as nonlinear springs along the pile length, where lateral soil pressure p varies with pile deflection y , allowing computation of pile deflection, moment, and shear under lateral load.

#435 · Foundations and Retaining Structures for Lateral Loads

Why are grade beams or foundation ties required between isolated footings in high seismic zones?

Answer

ASCE 7-16 Section 12.13.8 requires ties between individual pile caps and spread footings in SDC C and higher to resist differential lateral displacement, preventing footings from moving independently during ground shaking.

#436 · Foundations and Retaining Structures for Lateral Loads

What is the difference between free-head and fixed-head conditions for a laterally loaded pile?

Answer

A free-head pile can rotate freely at the pile cap connection under lateral load, while a fixed-head pile is restrained against rotation by the cap, resulting in different deflection, moment distribution, and reduced maximum deflection for fixed-head piles.

#437 · Foundations and Retaining Structures for Lateral Loads

How does group effect reduce lateral capacity in closely spaced pile groups?

Answer

Overlapping stress zones between adjacent piles reduce the soil resistance available to each pile compared to an isolated pile, requiring p-multiplier reduction factors applied to individual pile p-y curves within a group.

#438 · Foundations and Retaining Structures for Lateral Loads

What lateral pressure condition governs design of a rigid basement wall braced at both floor and top?

Answer

At-rest earth pressure, using K_0 , typically governs since the wall is restrained from the deflection needed to develop the lower active pressure state, resulting in higher design lateral pressure than an unrestrained cantilever wall.

#439 · Foundations and Retaining Structures for Lateral Loads

How does a sloped backfill surface affect the active earth pressure coefficient in Coulomb theory?

Answer

A backfill sloped upward away from the wall increases the active earth pressure coefficient compared to a level backfill, since the additional soil mass above the theoretical failure wedge increases the driving force on the wall.

#440 · Foundations and Retaining Structures for Lateral Loads

What is hydrostatic pressure and why must it be added to lateral earth pressure for a saturated backfill?

Answer

Hydrostatic pressure is the pressure exerted by groundwater behind a wall, increasing linearly with depth below the water table; it must be added to the effective lateral earth pressure since submerged soil unit weight is used for the soil component alone.

#441 · Foundations and Retaining Structures for Lateral Loads

Why is drainage behind a retaining wall critical to controlling design lateral pressure?

Answer

Without adequate drainage, water accumulates behind the wall and adds significant hydrostatic pressure on top of soil pressure, substantially increasing total lateral load beyond what the wall was designed to resist if drainage was assumed.

#442 · Foundations and Retaining Structures for Lateral Loads

What is global stability failure as distinct from sliding failure of a retaining wall?

Answer

Global stability failure is a deep-seated rotational slip surface passing beneath and beyond the wall and footing through the surrounding soil mass, which must be checked separately from local sliding along the footing base.

#443 · Foundations and Retaining Structures for Lateral Loads

How is the resultant location on a footing base determined for overturning and bearing checks?

Answer

The resultant location is found by summing moments of all vertical and lateral forces about a reference point, typically the footing centerline or toe, then dividing net moment by net vertical force to find the eccentricity.

#444 · Foundations and Retaining Structures for Lateral Loads

What is meant by 'net allowable bearing pressure' when checking footing capacity?

Answer

Net allowable bearing pressure is the gross allowable bearing capacity minus the overburden pressure of soil removed above the footing base, representing the additional pressure the footing can safely carry beyond the excavated soil weight.

#445 · Foundations and Retaining Structures for Lateral Loads

Why must foundation ties between adjacent pile caps also be checked for compression, not just tension?

Answer

Seismic demand can reverse direction, so ties must transmit force in either tension or compression between footings; a tie designed only for tension may buckle or provide no resistance when the load reverses to compression.

#446 · Foundations and Retaining Structures for Lateral Loads

What lateral load combination typically produces maximum sliding demand at a foundation?

Answer

The combination of full lateral wind or seismic force with minimum coincident dead load, since reduced vertical load lowers available friction resistance while the full horizontal shear demand remains unchanged.

#447 · Foundations and Retaining Structures for Lateral Loads

How does soil cohesion contribute to passive resistance calculations distinct from friction angle?

Answer

Cohesion provides an intercept term in the passive pressure equation independent of overburden depth, adding constant resistance from the soil's inherent shear strength, while friction angle governs the pressure component that increases with depth.

#448 · Foundations and Retaining Structures for Lateral Loads

What is a mat or raft foundation's advantage for resisting lateral overturning compared to isolated footings?

Answer

A mat foundation distributes vertical and overturning loads over a much larger contact area, reducing peak bearing pressure and increasing overturning resistance through the greater moment arm across the full mat width.

#449 · Foundations and Retaining Structures for Lateral Loads

Why is the point of fixity concept used in laterally loaded pile analysis?

Answer

The point of fixity is an equivalent depth below grade where a pile can be modeled as rigidly fixed for simplified analysis, replacing complex soil-structure interaction with a shorter effective cantilever length for hand calculations.

#450 · Foundations and Retaining Structures for Lateral Loads

What is the effect of an embedded footing depth on passive resistance available for sliding checks?

Answer

Greater embedment depth increases the height of soil available to develop passive pressure against the footing face, increasing total passive resistance force since passive pressure increases with the square of the embedded height.

#451 · Foundations and Retaining Structures for Lateral Loads

How should uplift on a foundation be checked when net tension develops under seismic overturning?

Answer

The foundation, its reinforcement, and its connection to the superstructure must resist net uplift tension directly, since spread footings without piles or tension anchors cannot resist tension and would separate from bearing soil.

#452 · Foundations and Retaining Structures for Lateral Loads

What is a batter pile and why is it used to resist lateral load?

Answer

A batter pile is installed at an angle rather than vertically, allowing it to resist lateral load axially through its inclined orientation rather than relying solely on lateral bending capacity of a vertical pile.

#453 · Foundations and Retaining Structures for Lateral Loads

Why can batter piles be undesirable in high seismic regions despite their lateral efficiency?

Answer

Batter piles can attract large axial forces during seismic shaking due to their stiff axial load path, potentially overloading the pile or cap connection, so many seismic codes restrict or discourage their use in high SDC zones.

#454 · Foundations and Retaining Structures for Lateral Loads

What is the purpose of checking eccentricity ratio e/B for a footing under combined loading?

Answer

The eccentricity ratio, eccentricity divided by footing width, determines whether the resultant falls within the middle third, governing whether the full base area remains in compression or whether an effective reduced area must be used.

#455 · Foundations and Retaining Structures for Lateral Loads

How does a deep foundation system change the overturning check compared to a shallow spread footing?

Answer

Deep foundations resist overturning primarily through axial tension and compression developed in individual piles at the perimeter of the pile group rather than through soil bearing pressure redistribution beneath a single footing.

#456 · Foundations and Retaining Structures for Lateral Loads

What role does the coefficient of friction between concrete and soil play in sliding resistance calculations?

Answer

It relates the vertical bearing force to available frictional sliding resistance at the footing base interface, with typical values around 0.3 to 0.5 depending on soil type, directly scaling the friction component of total resistance.

#457 · Foundations and Retaining Structures for Lateral Loads

Why must retaining wall stem design consider both the active earth pressure and any applied surcharge simultaneously?

Answer

The stem must resist the combined bending moment and shear from both soil-induced active pressure and any surcharge load acting concurrently, since ignoring either source would understate the total lateral demand on the wall stem.

#458 · Foundations and Retaining Structures for Lateral Loads

What is meant by 'toe pressure' and 'heel pressure' in cantilever retaining wall footing design?

Answer

Toe pressure is the bearing pressure at the footing edge nearest the retained soil face, and heel pressure is at the opposite edge under the backfill; both are computed from the resultant location to verify allowable bearing is not exceeded.

#459 · Foundations and Retaining Structures for Lateral Loads

How does seismic lateral earth pressure differ from static active pressure in retaining wall design?

Answer

Seismic lateral earth pressure adds a dynamic increment, often estimated using the Mononobe-Okabe method, that accounts for inertial soil forces during ground shaking, producing higher total pressure than the static active pressure alone.

#460 · Foundations and Retaining Structures for Lateral Loads

Why is a minimum embedment depth specified for footings resisting lateral load in frost-susceptible or expansive soils?

Answer

Adequate embedment ensures the footing bears below the depth affected by frost heave or seasonal moisture-driven soil volume change, preventing loss of bearing support or unintended uplift forces that would compromise lateral load resistance.

#461 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the purpose of a Tier 1 seismic evaluation for an existing building?

Answer

A Tier 1 screening evaluation uses simplified checklists of quick-check calculations to rapidly identify potential seismic deficiencies in an existing building, determining whether a more detailed Tier 2 or Tier 3 analysis is warranted.

#462 • General Analysis of Existing Secondary and Nonbuilding Structures

Define a soft story irregularity in an existing building.

Answer

A soft story exists when one story's lateral stiffness is significantly less than the story above, commonly less than 70 percent, often caused by tall floor heights or open ground-floor parking, concentrating drift and damage there.

#463 • General Analysis of Existing Secondary and Nonbuilding Structures

Why are soft story buildings particularly vulnerable during earthquakes?

Answer

Lateral drift concentrates almost entirely within the flexible story, causing large inelastic deformation demand on the limited lateral elements there, which can lead to story collapse before upper, stiffer stories experience significant damage.

#464 • General Analysis of Existing Secondary and Nonbuilding Structures

What is a weak story irregularity as distinct from a soft story?

Answer

A weak story exists when the lateral strength of one story, commonly less than 80 percent of the story above, is insufficient relative to demand, focusing on strength rather than stiffness, though the two often occur together.

#465 • General Analysis of Existing Secondary and Nonbuilding Structures

How does ASCE 7-16 Chapter 15 classify nonbuilding structures for seismic design?

Answer

Chapter 15 divides nonbuilding structures into those similar to buildings, using a modified form of the building provisions, and those not similar to buildings, such as tanks, stacks, and towers, using structure-specific design coefficients.

#466 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the design basis for a nonbuilding structure similar to a building, such as a pipe rack supported on a braced steel frame?

Answer

A pipe rack supported on a braced steel frame is treated as similar to a building under ASCE 7-16 Table 15.4-1, using the same building-type R , C_d , and Ω_{m0} values as the analogous building seismic force-resisting system (here, a steel braced frame), since its structural system closely parallels an ordinary building lateral system even though its occupancy is industrial.

#467 • General Analysis of Existing Secondary and Nonbuilding Structures

What is a nonstructural component per ASCE 7-16 Chapter 13, and give an example.

Answer

A nonstructural component is an architectural, mechanical, or electrical element attached to or housed within a building that is not part of the primary lateral or gravity system, such as a rooftop HVAC unit, ceiling system, or piping.

#468 • General Analysis of Existing Secondary and Nonbuilding Structures

How is the seismic design force for a nonstructural component generally calculated?

Answer

The component force uses the formula F_p based on the component's operating weight, an amplification factor a_p , a response modification factor R_p , the component importance factor I_p , and the height of attachment within the structure, per ASCE 7-16 Section 13.3.1.

#469 • General Analysis of Existing Secondary and Nonbuilding Structures

Why does the height of attachment within a building affect nonstructural component seismic force?

Answer

Floor accelerations amplify with height due to the building's dynamic response, so a component mounted higher in the structure experiences greater input acceleration and therefore requires a larger design force than one near the base.

#470 • General Analysis of Existing Secondary and Nonbuilding Structures

What is component importance factor I_p and when is it increased to 1.5?

Answer

I_p is increased from the default 1.0 to 1.5 for components in Risk Category IV structures, those containing hazardous materials, or life-safety components such as fire suppression systems needed for continued building function after an event.

#471 • General Analysis of Existing Secondary and Nonbuilding Structures

How does computer model verification through hand calculation serve as a quality check?

Answer

A simplified hand calculation of base shear, deflection, or member force provides an independent estimate to compare against computer output, catching gross modeling errors such as incorrect units, boundary conditions, or load application.

#472 • General Analysis of Existing Secondary and Nonbuilding Structures

What is a common first check when verifying a computer model's total seismic base shear?

Answer

Comparing the computer-calculated dynamic base shear against a hand-calculated equivalent lateral force base shear using the same seismic parameters confirms the model's total mass, period, and response spectrum inputs are reasonable.

#473 • General Analysis of Existing Secondary and Nonbuilding Structures

Why should reaction sums from a computer model be checked against total applied load?

Answer

Summing all support reactions and comparing to the total applied gravity or lateral load confirms global equilibrium in the model, revealing errors such as missing supports, incorrect load application, or unintended releases in the model.

#474 • General Analysis of Existing Secondary and Nonbuilding Structures

What retrofit strategy is commonly used to address a soft story deficiency in an existing wood-frame building?

Answer

Adding new plywood shear walls or steel moment frames at the deficient story increases lateral stiffness and strength there, reducing the stiffness discontinuity relative to the story above and redistributing drift demand more evenly.

#475 • General Analysis of Existing Secondary and Nonbuilding Structures

What is a vertical irregularity check for in-plane discontinuity of a lateral element?

Answer

It identifies where a shear wall or braced frame is offset horizontally from the wall or frame below it in the load path, creating an out-of-plane offset that requires the diaphragm and supporting elements to carry unintended transfer forces.

#476 • General Analysis of Existing Secondary and Nonbuilding Structures

Why are unreinforced masonry buildings frequently prioritized in existing building seismic mitigation programs?

Answer

Unreinforced masonry lacks ductile reinforcement to resist tension and out-of-plane bending, making it prone to brittle wall collapse and falling hazards during earthquake shaking, posing significant life-safety risk compared to reinforced systems.

#477 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the design intent behind seismic bracing requirements for suspended ceiling systems?

Answer

Lateral bracing wires and compression struts limit ceiling grid movement during shaking, preventing grid buckling, tile dislodgement, and detachment of the ceiling system from the supporting structure, which could otherwise fall and injure occupants.

#478 • General Analysis of Existing Secondary and Nonbuilding Structures

How does ASCE 7-16 treat seismic design of elevated water tanks compared to ground-supported tanks?

Answer

Elevated tanks are treated as nonbuilding structures not similar to buildings, using tower-specific R and C_d values that reflect the concentrated mass at height and limited ductility of the supporting tower structure, unlike ground-supported tanks.

#479 · General Analysis of Existing Secondary and Nonbuilding Structures

What is a torsional irregularity check used for in existing building seismic evaluation?

Answer

It compares the maximum story drift at one end of the building to the average drift, identifying whether the center of mass and center of rigidity are significantly offset, causing disproportionate twisting demand on perimeter elements.

#480 · General Analysis of Existing Secondary and Nonbuilding Structures

Why must anchorage of nonstructural equipment be verified in existing building seismic assessments?

Answer

Unanchored or inadequately anchored mechanical and electrical equipment can slide, overturn, or become a falling hazard during shaking, potentially damaging piping, causing fires, or injuring occupants even if the primary structure performs adequately.

#481 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the purpose of a rigid nonstructural component seismic force calculation versus a flexible one?

Answer

Rigid components with a fundamental period less than 0.06 seconds use a simplified force equation, while flexible components require accounting for dynamic amplification since they can resonate with building floor motion, increasing demand significantly.

#482 • General Analysis of Existing Secondary and Nonbuilding Structures

How is a computer model's mode shape and period output typically sanity-checked?

Answer

The calculated fundamental period is compared against an approximate period formula from ASCE 7-16 Section 12.8.2.1 based on building height and structure type, flagging significant discrepancies that may indicate modeling errors in mass or stiffness.

#483 • General Analysis of Existing Secondary and Nonbuilding Structures

What existing building condition commonly triggers a mandatory seismic retrofit under local ordinance?

Answer

Unreinforced masonry bearing wall buildings, non-ductile concrete frame buildings, and soft-story wood-frame buildings with multiple dwelling units are commonly targeted by local mandatory retrofit ordinances due to documented poor earthquake performance.

#484 • General Analysis of Existing Secondary and Nonbuilding Structures

What is meant by 'secondary structure' in the context of general lateral analysis for the SE exam?

Answer

A secondary structure is a smaller or ancillary structural element attached to or near the main building, such as an equipment platform, canopy, or screen wall, requiring its own lateral load evaluation independent of the main structure.

#485 • General Analysis of Existing Secondary and Nonbuilding Structures

Why is relative displacement compatibility checked for nonstructural components spanning between floors?

Answer

Components such as piping or partition walls that span vertically between floors must accommodate the differential story drift between attachment points without failure, so seismic relative displacement D_p must be calculated and detailed for.

#486 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the significance of ductility in evaluating an existing nonbuilding structure like a pipe rack?

Answer

Structures with limited ductile detailing, common in older pipe racks and industrial platforms, are assigned lower R values reflecting reduced capacity for inelastic deformation, resulting in higher required elastic design forces than ductile systems.

#487 • General Analysis of Existing Secondary and Nonbuilding Structures

How does the general lateral analysis domain address existing building diaphragm evaluation?

Answer

Existing diaphragms, especially unblocked wood sheathing or plain concrete slabs, are checked for adequate shear capacity and chord continuity, since older diaphragms often lack sufficient nailing, blocking, or continuous chords assumed in modern design.

#488 • General Analysis of Existing Secondary and Nonbuilding Structures

What is a common verification step for checking a finite element model's deflected shape under lateral load?

Answer

Visually comparing the deflected shape against expected behavior, such as confirming a cantilever shear wall deflects outward at the top with no unexpected kinks or reversals, helps catch incorrect boundary conditions or element connectivity errors.

#489 • General Analysis of Existing Secondary and Nonbuilding Structures

Why are freestanding equipment platforms treated as nonbuilding structures similar to buildings when they resemble a building frame?

Answer

When a platform has a lateral force resisting system comparable to buildings, such as braced or moment frames supporting equipment, ASCE 7-16 permits using building-like seismic provisions with platform-specific mass and configuration inputs.

#490 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the purpose of checking story drift limits during evaluation of an existing building's lateral system?

Answer

Comparing calculated story drift under design lateral load against code allowable drift limits identifies stories where excessive flexibility could cause nonstructural damage or, in severe cases, contribute to P-delta instability and collapse risk.

#491 • General Analysis of Existing Secondary and Nonbuilding Structures

How does P-delta effect become a concern in evaluating an existing soft story building?

Answer

Large lateral drift concentrated in a soft story combined with gravity load creates an additional secondary overturning moment, which can amplify drift further and destabilize the story if the lateral stiffness is insufficient to resist the compounding effect.

#492 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the design consideration for anchoring a rooftop mechanical unit against overturning?

Answer

The unit's anchorage must resist the overturning moment created by its seismic design force acting at its center of gravity relative to its base, checked against the resisting moment from unit self-weight and anchor tension capacity.

#493 • General Analysis of Existing Secondary and Nonbuilding Structures

Why is component amplification factor a_p higher for rigidly mounted equipment near the roof than for base-isolated equipment?

Answer

Rigid equipment near the roof directly tracks the amplified floor acceleration at that height with little additional filtering, while flexibly mounted or isolated equipment can reduce transmitted acceleration, though a_p accounts for the component's own dynamic response characteristics.

#494 • General Analysis of Existing Secondary and Nonbuilding Structures

What existing building feature commonly causes an out-of-plane wall anchorage deficiency in tilt-up construction?

Answer

Older tilt-up buildings often used wood ledger connections relying on cross-grain bending to transfer out-of-plane wall forces to the roof diaphragm, a connection type now recognized as unreliable and commonly targeted for seismic retrofit.

#495 • General Analysis of Existing Secondary and Nonbuilding Structures

How is a computer model's applied wind load typically verified for reasonableness?

Answer

The total wind base shear from the model is compared against a hand calculation using the velocity pressure equation and appropriate exposure and gust factors from ASCE 7-16 Chapter 26, confirming consistent input parameters and projected areas.

#496 • General Analysis of Existing Secondary and Nonbuilding Structures

What is the purpose of evaluating an existing building's redundancy when performing a seismic assessment?

Answer

Limited redundancy, such as a single line of lateral resistance in one direction, means the loss of one element could cause disproportionate structural distress, informing whether targeted strengthening of critical elements is warranted during evaluation.

#497 • General Analysis of Existing Secondary and Nonbuilding Structures

What load path issue is commonly found when evaluating diaphragm continuity at re-entrant corners in existing buildings?

Answer

Re-entrant corners create stress concentrations from the geometric discontinuity, and older buildings often lack adequate collector reinforcement or drag struts at these locations, risking local diaphragm tearing during strong shaking.

#498 • General Analysis of Existing Secondary and Nonbuilding Structures

Why must a nonbuilding structure supported on top of another building, such as a rooftop antenna tower, be evaluated for both its own dynamics and interaction with the supporting structure?

Answer

The supported structure's mass and stiffness can interact with the main building's dynamic response, potentially amplifying forces beyond what either structure would experience independently, requiring combined or conservatively enveloped analysis per ASCE 7-16.

#499 • General Analysis of Existing Secondary and Nonbuilding Structures

What is a reasonable first check when a computer model reports an unexpectedly low fundamental period for a tall building?

Answer

Checking for unintended rigid connections, missing releases, or excessive assumed stiffness in the model, since an unrealistically short period suggests the model is stiffer than the actual structure, understating dynamic seismic demand.

#500 • General Analysis of Existing Secondary and Nonbuilding Structures

Why is verifying total seismic weight W a critical hand-check step for any lateral analysis computer model?

Answer

Seismic base shear is directly proportional to effective seismic weight, so an error in mass input, such as missing partition load or double-counted floor mass, propagates directly and proportionally into every calculated lateral force in the model.